

Samba AD in ubuntu 24.04

Samba Active Directory (AD) is a free, open-source implementation of Microsoft's Active Directory Domain Services (AD DS) that runs on Linux. It allows a Linux server to act as an AD Domain Controller (DC), offering centralized identity management, user authentication, and Group Policy management for Windows and Linux networks, providing a cost-effective alternative to Windows Server.

Follow the steps:

Install Samba AD in Ubuntu

Step-1: Set static IP on server (If required)

```
# ifconfig
```

```
# sudo cat /etc/netplan/50-cloud-init.yaml -----> Make changes in this file.
```

```
network:
  version: 2
  renderer: networkd
  ethernets:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.152.129/24
      gateway4: 192.168.152.2
      nameservers:
        addresses: [8.8.8.8, 1.1.1.1]
```

Eg:

Step-2: Change hostname (If required)

```
# hostname -f
```

```
# hostnamectl set-hostname master
```

```
siddharrth@freshers-server-2:~$ hostnamectl set-hostname master
==== AUTHENTICATING FOR org.freedesktop.hostname1.set-static-hostname ====
Authentication is required to set the statically configured local hostname, as well as the pretty hostname.
Authenticating as: siddharrth
Password:
==== AUTHENTICATION COMPLETE ====
siddharrth@freshers-server-2:~$ bash
siddharrth@master:~$ hostname -f
master
```

Step-3: modify the hostname in host file

sudo vi /etc/hosts (Ensure the entry)

```
siddharrth@master:~$ cat /etc/hosts
127.0.0.1 localhost
10.10.40.88 master.smb.com      master
# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
siddharrth@master:~$ hostname -f
master.smb.com
```

Replace IP and hostname like this

Step-4: modify resolv.conf

sudo vi /etc/resolv.conf (Ensure the entry)

nameserver 127.0.0.1

options edns0 trust-ad

search smb.com

```
siddharrth@master:~$ sudo vi /etc/resolv.conf
nameserver 127.0.0.1
options edns0 trust-ad
search smb.com
```

```
siddharrth@master:~$ ping master.smb.com
PING master.smb.com (10.10.40.88) 56(84) bytes of data.
64 bytes from master.smb.com (10.10.40.88): icmp_seq=1 ttl=64 time=0.032 ms
64 bytes from master.smb.com (10.10.40.88): icmp_seq=2 ttl=64 time=0.035 ms
64 bytes from master.smb.com (10.10.40.88): icmp_seq=3 ttl=64 time=0.034 ms
64 bytes from master.smb.com (10.10.40.88): icmp_seq=4 ttl=64 time=0.056 ms
```

Step-5: Update the repo

sudo apt update

sudo apt upgrade

```
siddharrth@master:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
30 packages can be upgraded. Run 'apt list --upgradable' to see them.
siddharrth@master:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  apparmor libapparmor1 libnss-systemd libpam-systemd libsystemd-shared libsystemd0 libudev1 systemd systemd-dev
  systemd-resolved systemd-sysv systemd-timesyncd udev
The following packages will be upgraded:
  cloud-init coreutils fwupd libfwupd2 libnetplan1 libnftables1 linux-base lshw netplan-generator netplan.io nftables
  python3-netplan python3-software-properties software-properties-common sosreport systemd-hwe-hwdb
  ubuntu-drivers-common
17 upgraded, 0 newly installed, 0 to remove and 13 not upgraded.
Need to get 8,334 kB of archives.
After this operation, 285 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Step-6: Install the samba ad and dependencies

```
# sudo apt install samba winbind krb5-user krb5-config smbclient dnsutils acl attr libpam-winbind libnss-winbind
```

```
siddharrth@master:~$ sudo apt install samba winbind krb5-user krb5-config smbclient dnsutils acl attr libpam-winbind libnss-winbind
[sudo] password for siddharrth:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libavahi-client3 libavahi-common-data libavahi-common3 libboost-iostreams1.83.0 libboost-locale1.83.0
  libboost-thread1.83.0 libcephfs2 libcups2t64 libgssrpc4t64 libkadm5clnt-mit12 libkadm5srv-mit12 libkdb5-10t64
  libldb2 librados2 librdmacm1t64 libsmbclient0 libtalloc2 libtdb1 libtevent0t64 liburing2 libwbclient0
  python3-dnspython python3-gpg python3-ldb python3-markdown python3-samba python3-talloc python3-tdb
  samba-ad-provision samba-common samba-common-bin samba-dsdb-modules samba-libs samba-vfs-modules tdb-tools
Suggested packages:
  krb5-k5tls cups-common krb5-doc python3-trio python3-aiioquic python3-h2 python3-httpx python3-httpcore
  python-markdown-doc bind9 bind9utils ctdb ldb-tools ntp | chrony heimdal-clients cifs-utils
The following NEW packages will be installed:
  acl attr dnsutils krb5-config krb5-user libavahi-client3 libavahi-common-data libavahi-common3
  libboost-iostreams1.83.0 libboost-locale1.83.0 libboost-thread1.83.0 libcephfs2 libcups2t64 libgssrpc4t64
  libkadm5clnt-mit12 libkadm5srv-mit12 libkdb5-10t64 libldb2 libnss-winbind libpam-winbind librados2 librdmacm1t64
  libsmbclient0 libtalloc2 libtdb1 libtevent0t64 liburing2 libwbclient0 python3-dnspython python3-gpg python3-ldb
  python3-markdown python3-samba python3-talloc python3-tdb samba samba-ad-provision samba-common samba-common-bin
  samba-dsdb-modules samba-libs samba-vfs-modules smbclient tdb-tools winbind
0 upgraded, 45 newly installed, 0 to remove and 0 not upgraded.
Need to get 21.2 MB of archives.
After this operation, 111 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Step-7: Complete the samba configuration setup:

Package configuration

Configuring Kerberos Authentication

When users attempt to use Kerberos and specify a principal or user name without specifying what administrative Kerberos realm that principal belongs to, the system appends the default realm. The default realm may also be used as the realm of a Kerberos service running on the local machine. Often, the default realm is the uppercase version of the local DNS domain.

Default Kerberos version 5 realm:

SMB.COM

<Ok>

Package configuration

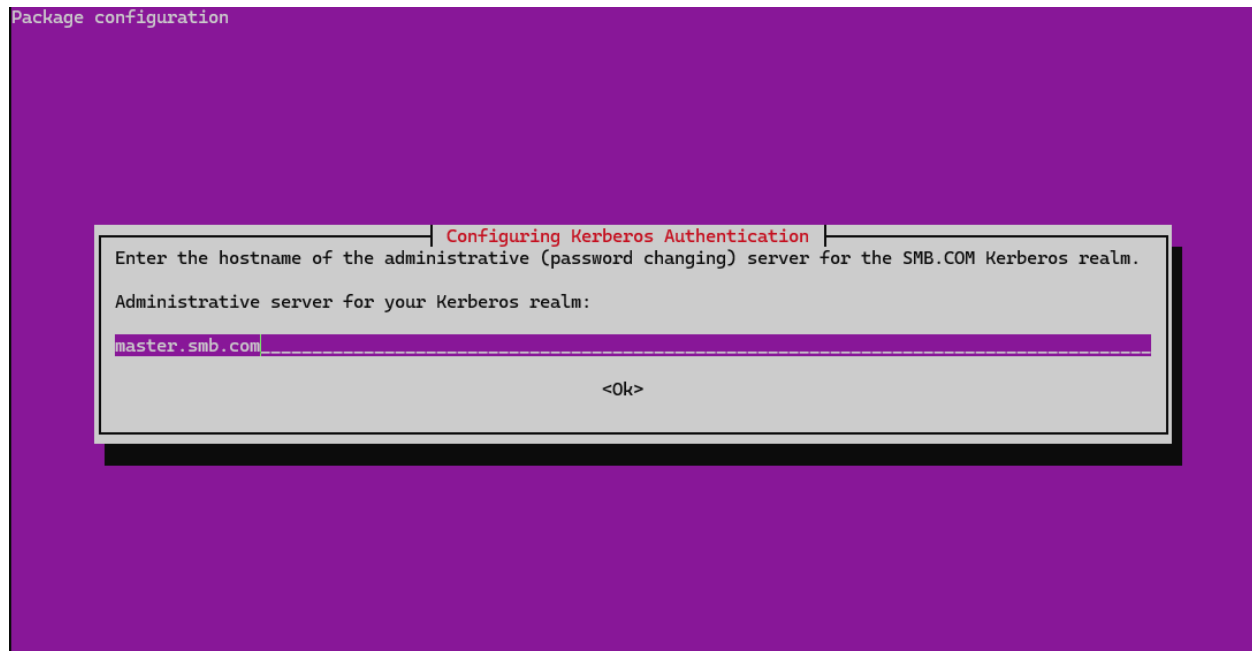
Configuring Kerberos Authentication

Enter the hostnames of Kerberos servers in the SMB.COM Kerberos realm separated by spaces.

Kerberos servers for your realm:

master.smb.com

<Ok>



Step-8: Stop/Disable samba-ad-dc Services

```
# sudo systemctl stop samba-ad-dc.service smb.service nmbd.service winbind.service
```

```
# sudo systemctl disable samba-ad-dc.service smb.service nmbd.service winbind.service
```

```
siddharrth@master:~$ sudo systemctl stop samba-ad-dc.service smb.service nmbd.service winbind.service
siddharrth@master:~$ sudo systemctl disable samba-ad-dc.service smb.service nmbd.service winbind.service
Synchronizing state of samba-ad-dc.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable samba-ad-dc
Synchronizing state of nmbd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable nmbd
Synchronizing state of winbind.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable winbind
Removed "/etc/systemd/system/smb.service".
Removed "/etc/systemd/system/nmb.service".
Removed "/etc/systemd/system/samba.service".
Removed "/etc/systemd/system/multi-user.target.wants/winbind.service".
Removed "/etc/systemd/system/multi-user.target.wants/nmbd.service".
Removed "/etc/systemd/system/multi-user.target.wants/smbd.service".
Removed "/etc/systemd/system/multi-user.target.wants/samba-ad-dc.service".
```

Step-9: Provision the domain

```
# sudo samba-tool domain provision --use-rfc2307 --interactive
```

Note: Hit Enter for all expect admin password: Optit@123

```
siddharrth@master:~$ sudo samba-tool domain provision --use-rfc2307 --interactive
Realm [SMB.COM]:
Domain [SMB]:
Server Role (dc, member, standalone) [dc]:
DNS backend (SAMBA_INTERNAL, BIND9_FLATFILE, BIND9_DLZ, NONE) [SAMBA_INTERNAL]:
DNS forwarder IP address (write 'none' to disable forwarding) [127.0.0.53]: 8.8.8.8
Administrator password:
Retype password:
INFO 2026-04-10 06:24:10,740 pid:2696 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2128: Looking up IPv4
addresses
INFO 2026-04-10 06:24:10,740 pid:2696 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2145: Looking up IPv6
addresses
WARNING 2026-04-10 06:24:10,741 pid:2696 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2152: No IPv6 addre
ss will be assigned
INFO 2026-04-10 06:24:11,102 pid:2696 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2318: Setting up share
.ldb
INFO 2026-04-10 06:24:11,201 pid:2696 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2322: Setting up secre
ts.ldb
INFO 2026-04-10 06:24:11,285 pid:2696 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2327: Setting up the r
egistry
INFO 2026-04-10 06:24:11,577 pid:2696 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2330: Setting up the p
rivileges database
```

Annotations in the terminal image:
 - Red box around the command: `sudo samba-tool domain provision --use-rfc2307 --interactive`
 - Red box around the prompts: `Realm [SMB.COM]:`, `Domain [SMB]:`, `Server Role (dc, member, standalone) [dc]:`, `DNS backend (SAMBA_INTERNAL, BIND9_FLATFILE, BIND9_DLZ, NONE) [SAMBA_INTERNAL]:`, `DNS forwarder IP address (write 'none' to disable forwarding) [127.0.0.53]:`
 - Red box around the password prompts: `Administrator password:`, `Retype password:`
 - Red arrow pointing to the first prompt: `<ENTER> go with default.`
 - Red arrow pointing to the IP address: `Enter IP`
 - Red arrow pointing to the password prompts: `Enter the Admin Password`

Step-10: Move and create link of krb5.conf

```
# sudo mv /etc/krb5.conf /etc/krb5.conf.bkp

# sudo ln -s /var/lib/samba/private/krb5.conf /etc/
```

```
siddharrth@master:~$ sudo mv /etc/krb5.conf /etc/krb5.conf.bkp
[sudo] password for siddharrth:
siddharrth@master:~$ ls /etc/krb5.conf
ls: cannot access '/etc/krb5.conf': No such file or directory
siddharrth@master:~$ sudo ln -s /var/lib/samba/private/krb5.conf /etc/
lrwxrwxrwx 1 root root 32 Apr 10 07:03 krb5.conf -> /var/lib/samba/private/krb5.conf
```

Step-11: Unmask the Samba service

```
# sudo systemctl unmask samba-ad-dc.service

# sudo systemctl start samba-ad-dc.service

# sudo systemctl enable samba-ad-dc.service

# sudo systemctl status samba-ad-dc.service
```

```
siddharrth@master:~$ sudo systemctl unmask samba-ad-dc.service
siddharrth@master:~$ sudo systemctl start samba-ad-dc.service
```

Annotations in the terminal image:
 - Red arrow pointing to the first command: `unmask and start the samba`
 - Red arrow pointing to the second command: `unmask and start the samba`

```

siddharrth@master:~$ sudo systemctl enable samba-ad-dc.service ← Enable the samba
Synchronizing state of samba-ad-dc.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable samba-ad-dc
Created symlink /etc/systemd/system/samba.service → /usr/lib/systemd/system/samba-ad-dc.service.
Created symlink /etc/systemd/system/multi-user.target.wants/samba-ad-dc.service → /usr/lib/systemd/system/samba-ad-dc.service.
siddharrth@master:~$ sudo systemctl status samba-ad-dc.service ← Status of the service
● samba-ad-dc.service - Samba AD Daemon
   Loaded: loaded (/usr/lib/systemd/system/samba-ad-dc.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-04-10 07:11:37 UTC; 48s ago
     Docs: man:samba(8)
           man:samba(7)
           man:smb.conf(5)
  Main PID: 2950 (samba)
    Status: "samba: ready to serve connections..."
     Tasks: 58 (limit: 2263)
  Memory: 179.7M (peak: 258.4M)
       CPU: 8.120s
   CGroup: /system.slice/samba-ad-dc.service
           └─2950 "samba: root process"
             └─2952 "samba: tfork waiter process(2953)"
               └─2953 "samba: task[s3fs] pre-fork master"
                 └─2954 "samba: tfork waiter process(2956)"
                   └─2955 "samba: tfork waiter process(2957)"

```

Step-12: Ping tests on the domain and host

ping master.smb.com

ping master

ping google.com

host -t A master.smb.com

host -t A master

host -t SRV _ldap._tcp.smb.com

host -t SRV _kerberos._udp.smb.com

samba-tool domain info 127.0.0.1

```

siddharrth@master:~$ ping master.smb.com
PING master.smb.com (10.10.40.88) 56(84) bytes of data.
64 bytes from master.smb.com (10.10.40.88): icmp_seq=1 ttl=64 time=0.060 ms
64 bytes from master.smb.com (10.10.40.88): icmp_seq=2 ttl=64 time=0.037 ms
64 bytes from master.smb.com (10.10.40.88): icmp_seq=3 ttl=64 time=0.038 ms
64 bytes from master.smb.com (10.10.40.88): icmp_seq=4 ttl=64 time=0.054 ms
^C
--- master.smb.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3054ms
rtt min/avg/max/mdev = 0.037/0.047/0.060/0.010 ms
siddharrth@master:~$ ping google.com
PING google.com (142.251.43.46) 56(84) bytes of data.
64 bytes from pnmaaa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=1 ttl=115 time=6.28 ms
64 bytes from pnmaaa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=2 ttl=115 time=6.37 ms
64 bytes from pnmaaa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=3 ttl=115 time=6.22 ms
64 bytes from pnmaaa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=4 ttl=115 time=6.25 ms
^C
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 6.223/6.280/6.366/0.053 ms
siddharrth@master:~$ ping master
PING master.smb.com (10.10.40.88) 56(84) bytes of data.
64 bytes from master.smb.com (10.10.40.88): icmp_seq=1 ttl=64 time=0.022 ms
64 bytes from master.smb.com (10.10.40.88): icmp_seq=2 ttl=64 time=0.049 ms
64 bytes from master.smb.com (10.10.40.88): icmp_seq=3 ttl=64 time=0.044 ms
^C
--- master.smb.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2080ms
rtt min/avg/max/mdev = 0.022/0.038/0.049/0.011 ms

```

```

siddharrth@master:~$ host -t A master.smb.com
master.smb.com has address 10.10.40.88
siddharrth@master:~$ host -t A master
master is an alias for master.smb.com.
master.smb.com has address 10.10.40.88

```

```

siddharrth@master:~$ sudo samba-tool domain level show
Domain and forest function level for domain 'DC=smb,DC=com'

Forest function level: (Windows) 2008 R2
Domain function level: (Windows) 2008 R2
Lowest function level of a DC: (Windows) 2008 R2

```

```

siddharrth@master:~$ host -t SRV _ldap._tcp.smb.com
_ldap._tcp.smb.com has SRV record 0 100 389 master.smb.com.
siddharrth@master:~$ host -t SRV _kerberos._udp.smb.com
_kerberos._udp.smb.com has SRV record 0 100 88 master.smb.com.

```

```

siddharrth@master:~$ samba-tool domain info 127.0.0.1
Forest           : smb.com
Domain           : smb.com
Netbios domain   : SMB
DC name          : master.smb.com
DC netbios name  : MASTER
Server site      : Default-First-Site-Name
Client site      : Default-First-Site-Name
siddharrth@master:~$

```


Step-13: Verification of Kerberos authentication

kinit Administrator

Admin password: Optit@123

```
siddharrth@master:~$ kinit Administrator
Password for Administrator@SMB.COM:
Warning: Your password will expire in 41 days on Fri 22 May 2026 06:24:19 AM UTC
```

note: if you get error this  (Troubleshoot)

```
siddharrth@master:~$ kinit Administrator
kinit: Cannot find KDC for realm "SMB.COM" while getting initial credentials
```

Do the follow this step:

sudo vi /etc/krb5.conf

Add hostname or IP address in this file

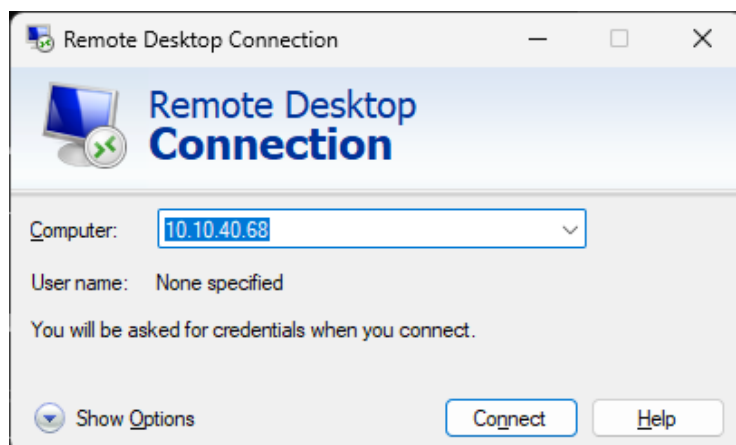
```
[libdefaults]
    default_realm = SMB.COM
    dns_lookup_realm = false
    dns_lookup_kdc = true

[realms]
SMB.COM = {
    kdc = master.smb.com
    default_domain = smb.com
}

[domain_realm]
    master = SMB.COM
```

Step-14: Now, Open windows machine for setup Samba AD

- Open RDP (Remote Desktop Protocol)



Windows Security

×

Enter your credentials

These credentials will be used to connect to 10.10.40.68.

Siddharrth G M

Password

OPTIT\siddharrth.gm

☐ Remember me

Click on 'More Choices'

Select different account

More choices

Siddharrth G M
OPTIT\siddharrth.gm

Use a different account

OK

Cancel

Windows Security

×

Enter your credentials

These credentials will be used to connect to 10.10.40.68.

User name

itadmin

Password

•••••

Domain: OPTIT

☐ Remember me

Enter username and password

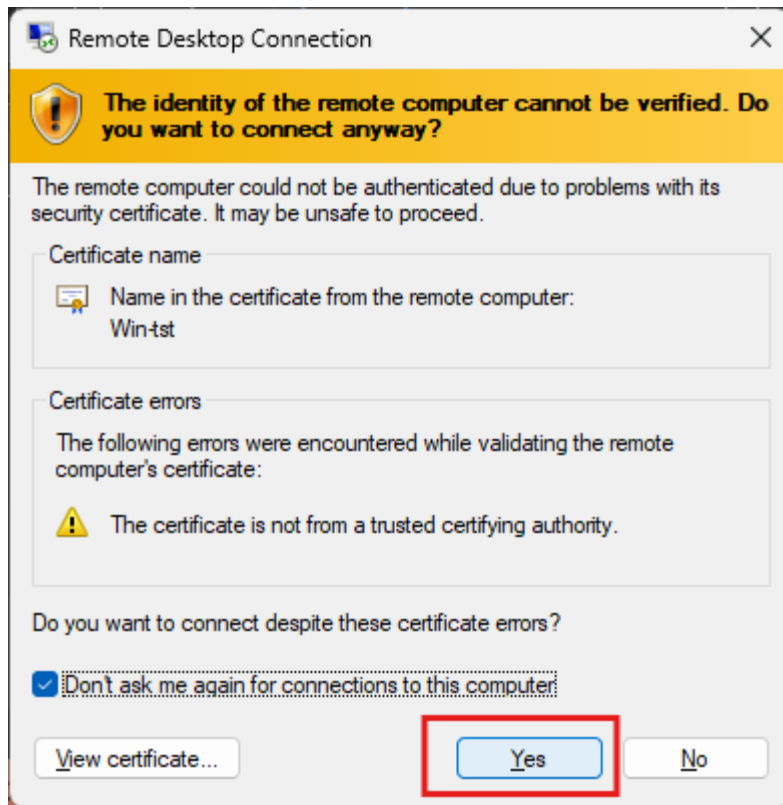
More choices

Siddharrth G M
OPTIT\siddharrth.gm

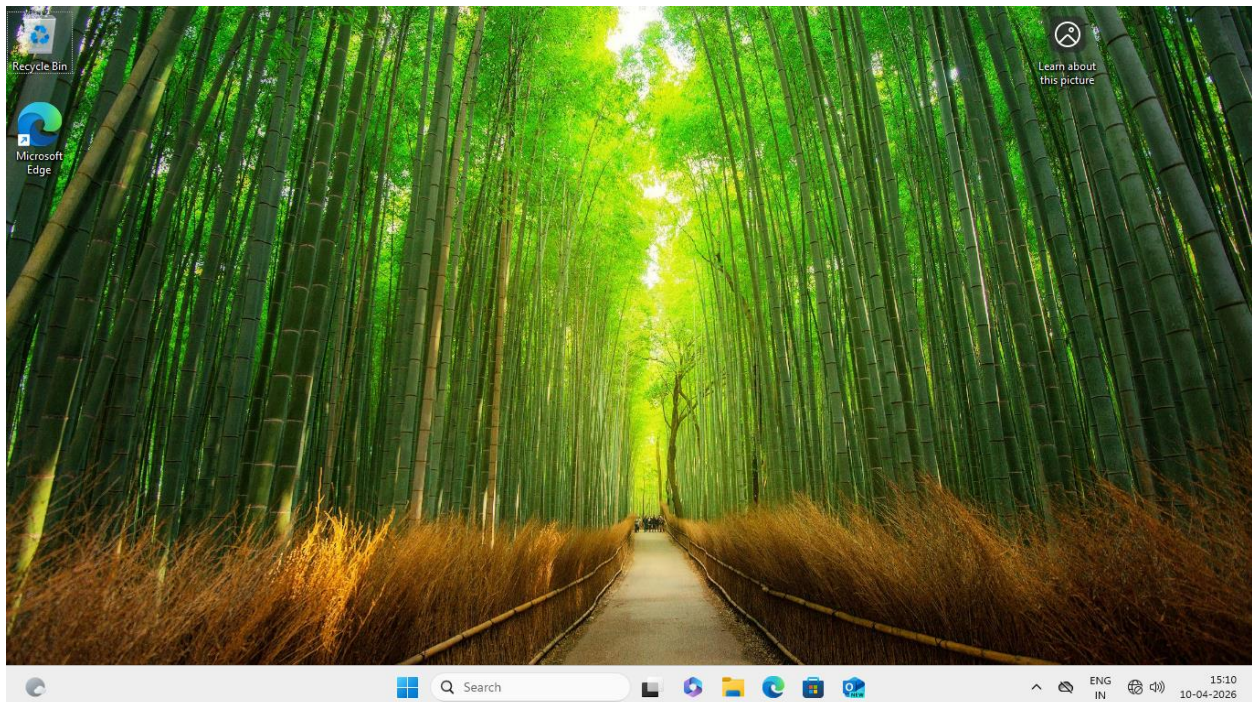
Use a different account

OK

Cancel

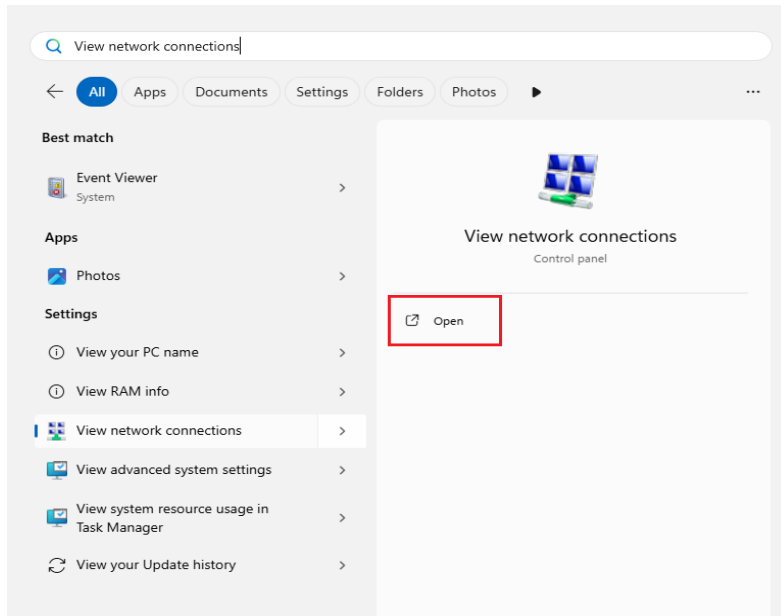


(Click yes)

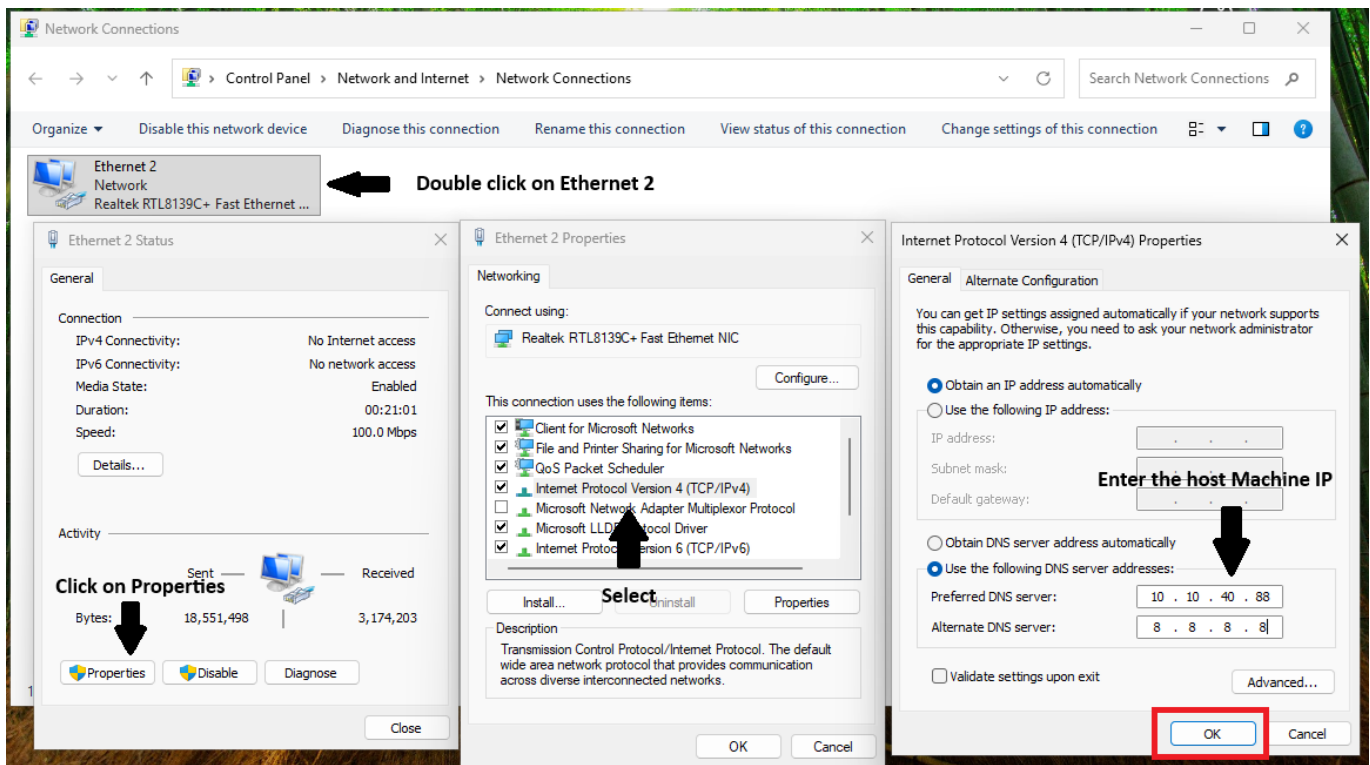


Step-15: Check the DNS in network settings

- Click 'windows' key ----> type 'view network connections'



- Double click on 'Ethernet 2' ----> Select 'Internet protocol version 4 (TCP/IPv6)'
- Select DNS server address ratio button ----> Enter the DNS address
 - Preferred DNS server: 10.10.40.88 -----> Admin machine IP address
 - Alternative DNS server: 8.8.8.8 -----> Alternative address (Google.com)
- Apply changes



Step- 16: Try ping the host machine from windows

- Open Command Prompt

ping master

ping 10.10.40.88

```
C:\Users\itadmin: ping 10.10.40.88

Pinging 10.10.40.88 with 32 bytes of data:
Reply from 10.10.40.88: bytes=32 time<1ms TTL=64
Reply from 10.10.40.88: bytes=32 time<1ms TTL=64
Reply from 10.10.40.88: bytes=32 time<1ms TTL=64
Reply from 10.10.40.88: bytes=32 time<1ms TTL=64

Ping statistics for 10.10.40.88:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

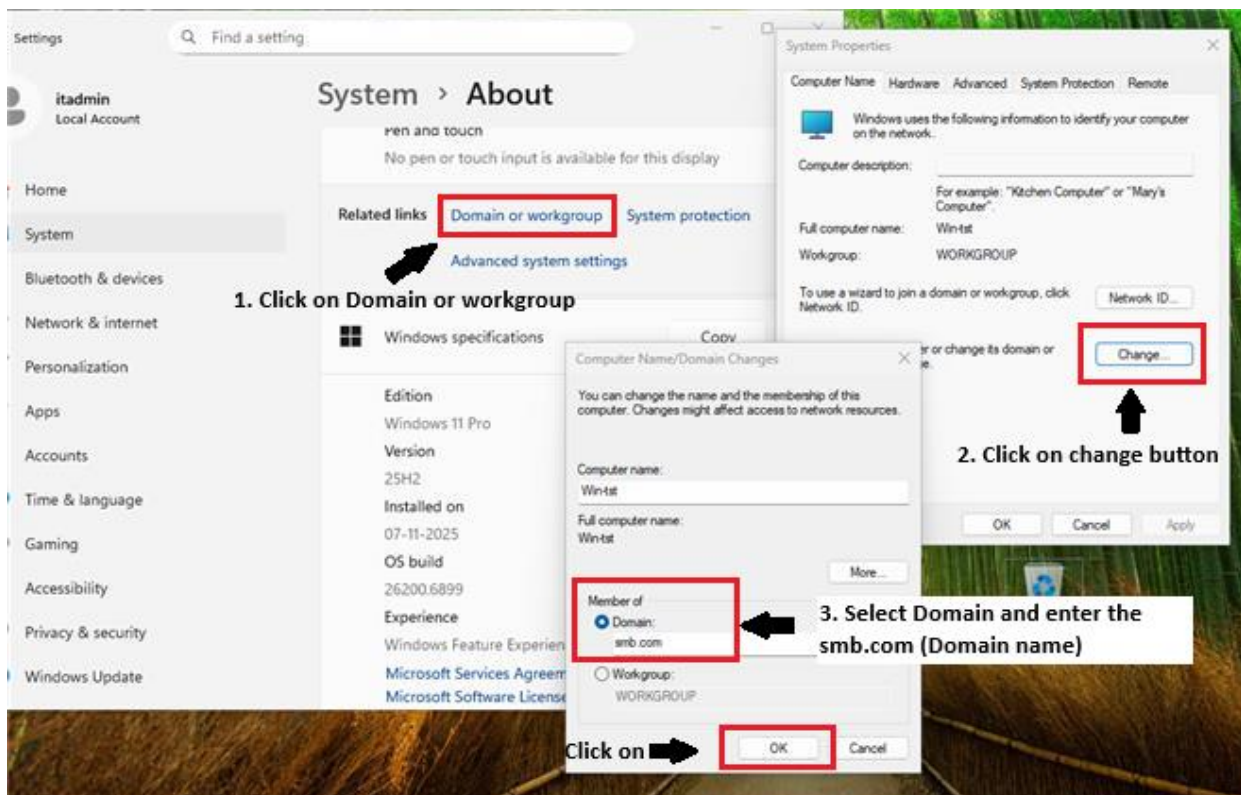
C:\Users\itadmin: ping master

Pinging master [10.10.40.88] with 32 bytes of data:
Reply from 10.10.40.88: bytes=32 time<1ms TTL=64
Reply from 10.10.40.88: bytes=32 time<1ms TTL=64
Reply from 10.10.40.88: bytes=32 time<1ms TTL=64
Reply from 10.10.40.88: bytes=32 time<1ms TTL=64

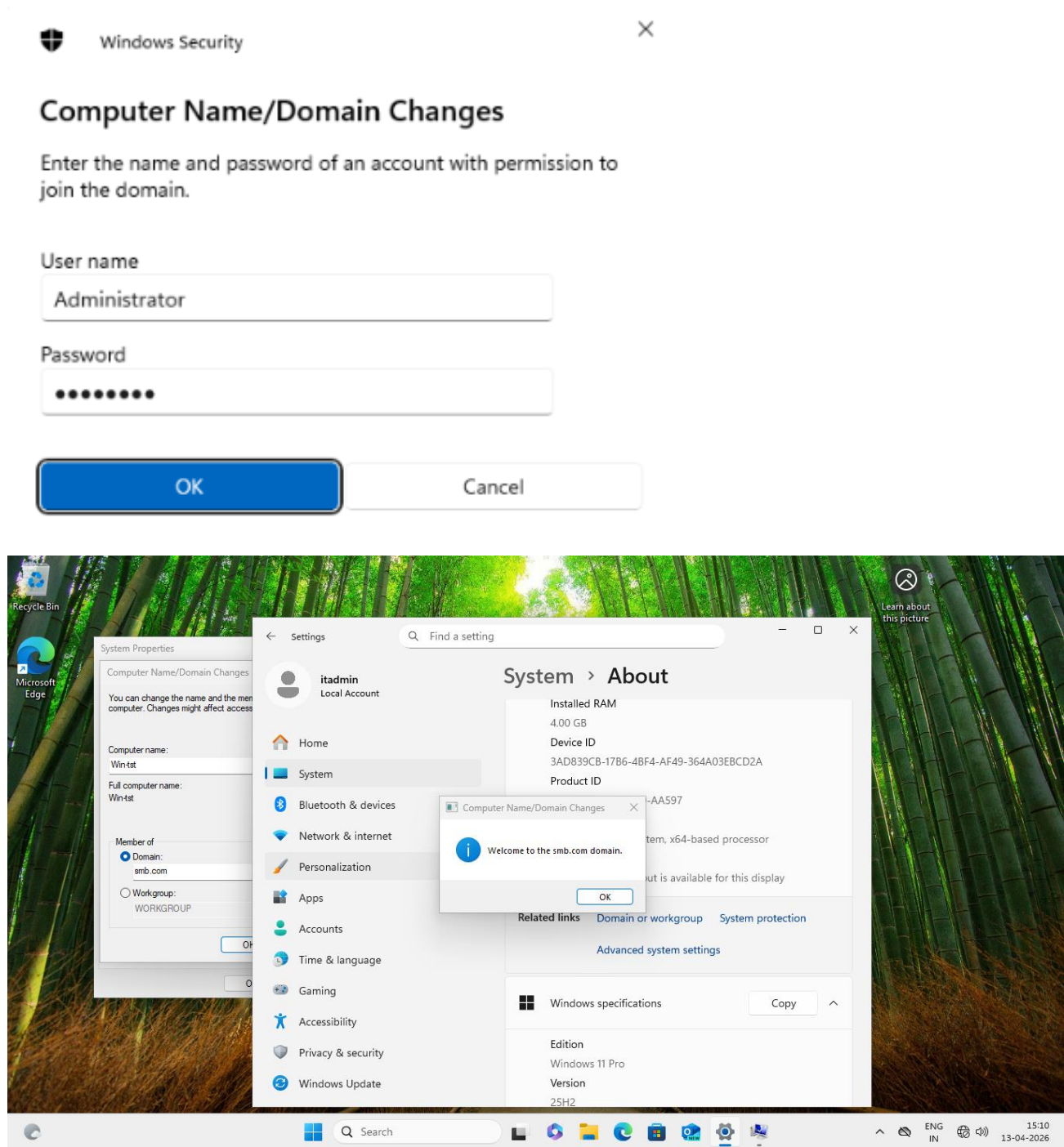
Ping statistics for 10.10.40.88:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Step-17: Change the domain

- Open setting ---> 'computer name' select domain 'change' button
- Select the Domain radio option in Member of field "Domain"
- Enter the Domain name and click ok



- Enter the Domain Username and Password



- Restart the system to reflect the changes

Step-18: Create a new user in AD (in Ubuntu CLI)

```
# sudo samba-tool user add sid
```

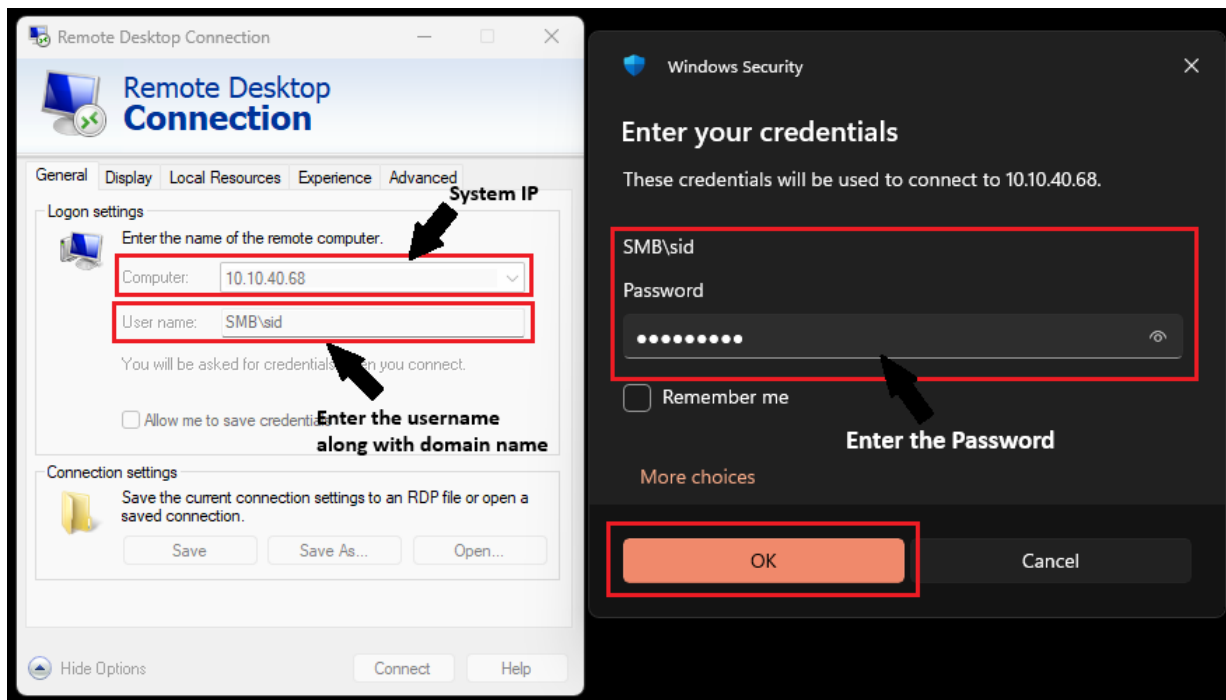
```
siddharrth@master:~$ sudo !!
sudo samba-tool user add sid
[sudo] password for siddharrth:
New Password:
Retype Password:
User 'sid' added successfully
```

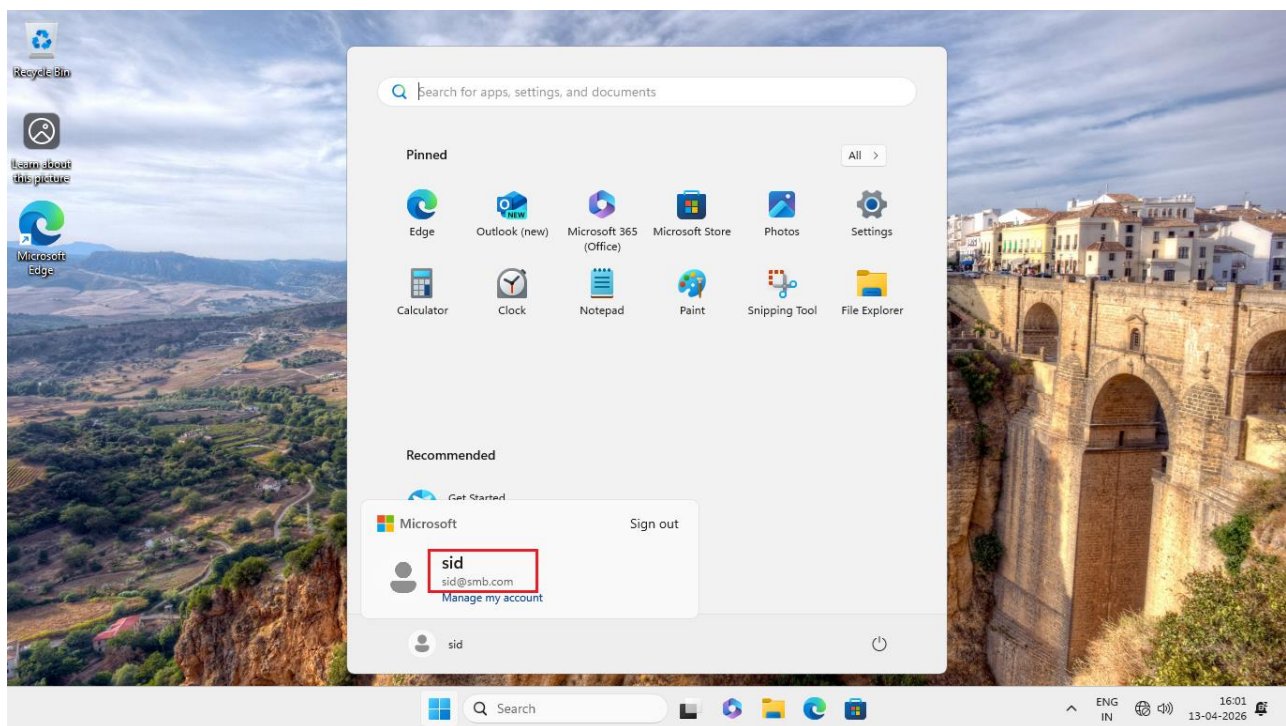
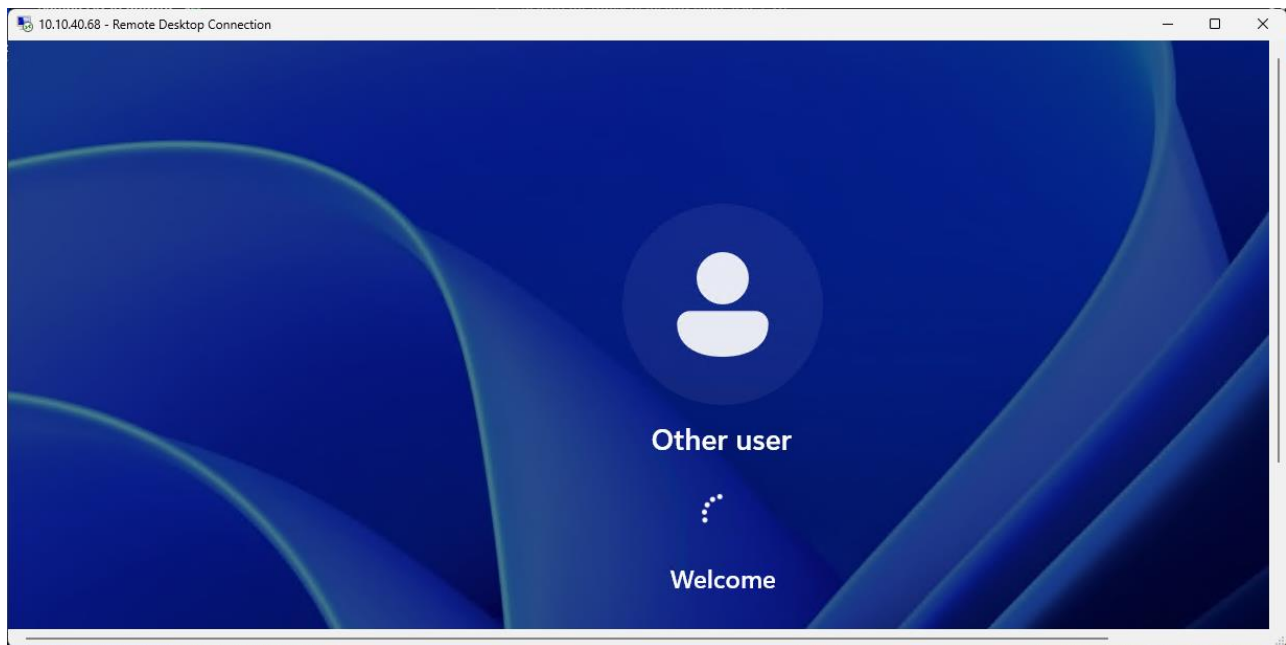
sudo samba-tool domain passwordsettings show

```
siddharrth@master:~$ sudo samba-tool domain passwordsettings show
Password information for domain 'DC=smb,DC=com'

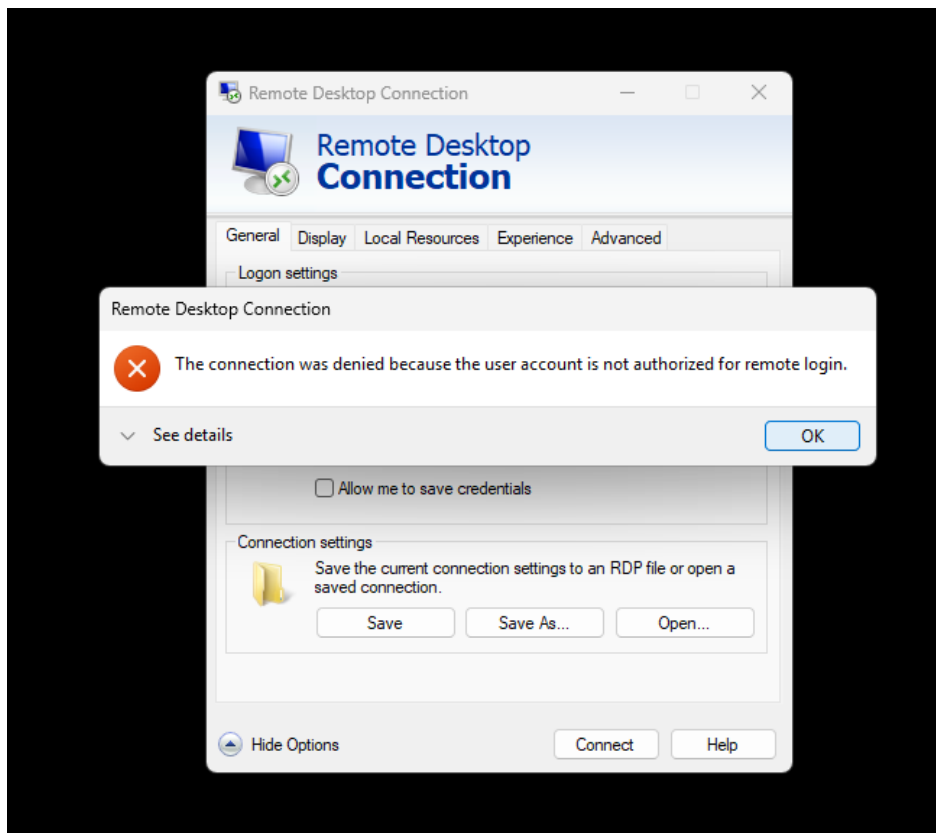
Password complexity: on
Store plaintext passwords: off
Password history length: 24
Minimum password length: 7
Minimum password age (days): 1
Maximum password age (days): 42
Account lockout duration (mins): 30
Account lockout threshold (attempts): 0
Reset account lockout after (mins): 30
```

Step-19: Try login the Windows Machine



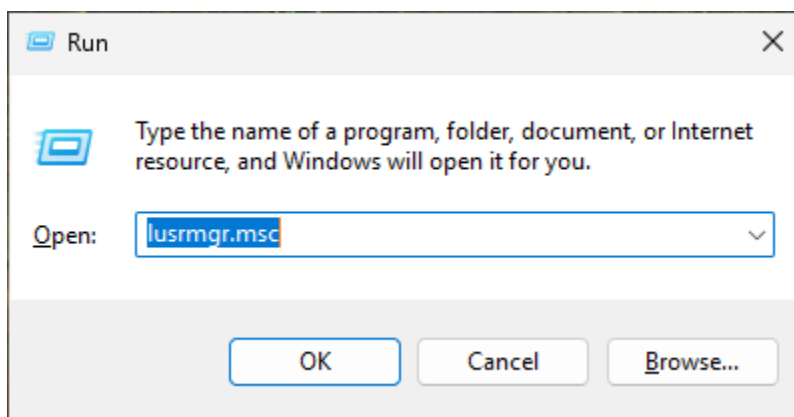


- It is login into the smb domain in sid user.
- If, Getting error in login RDP connection like this

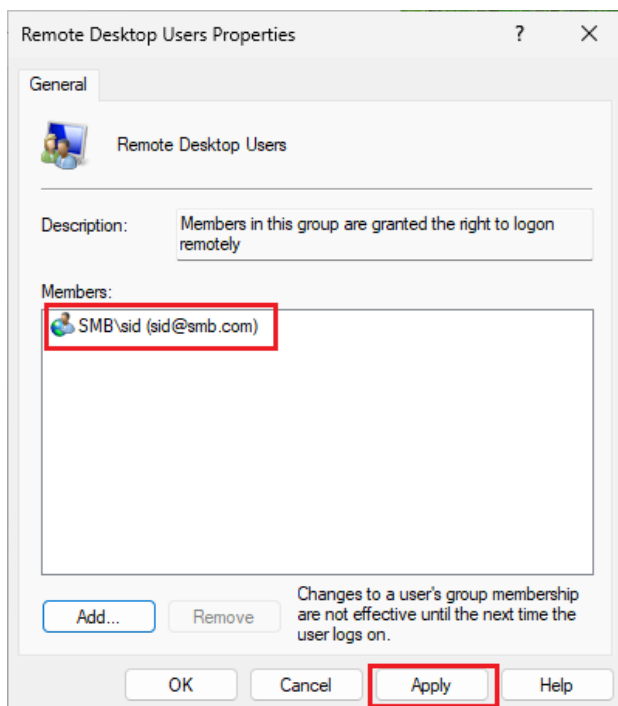
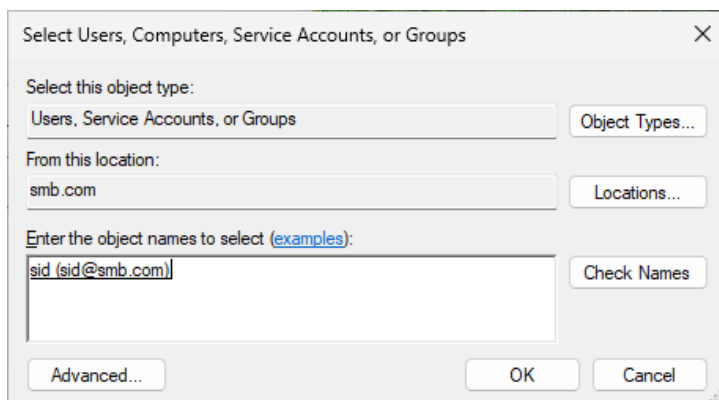
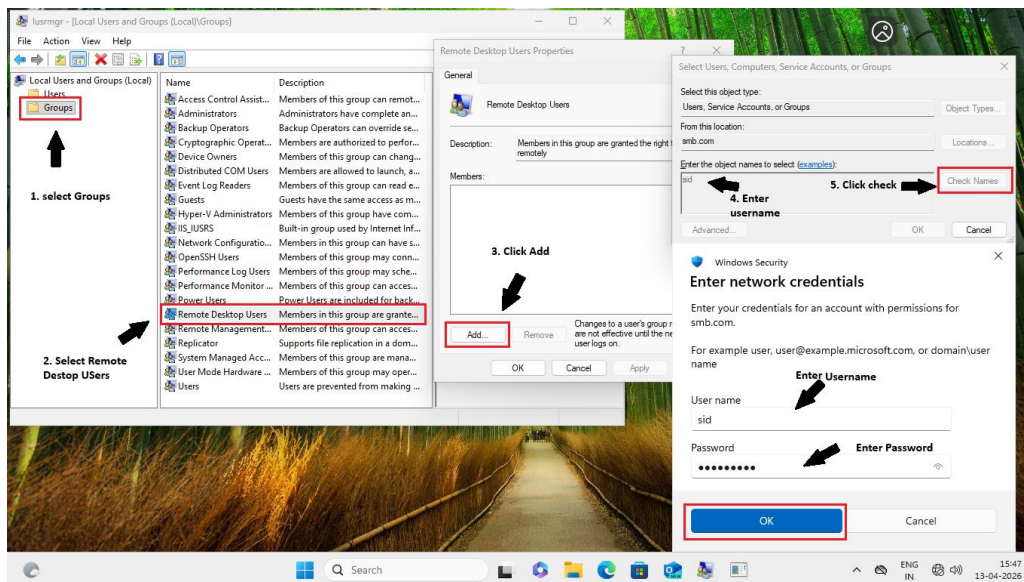


Follow this steps:

- Open run (win + r) -> Type lusrmgr.msc (Local users and groups management)



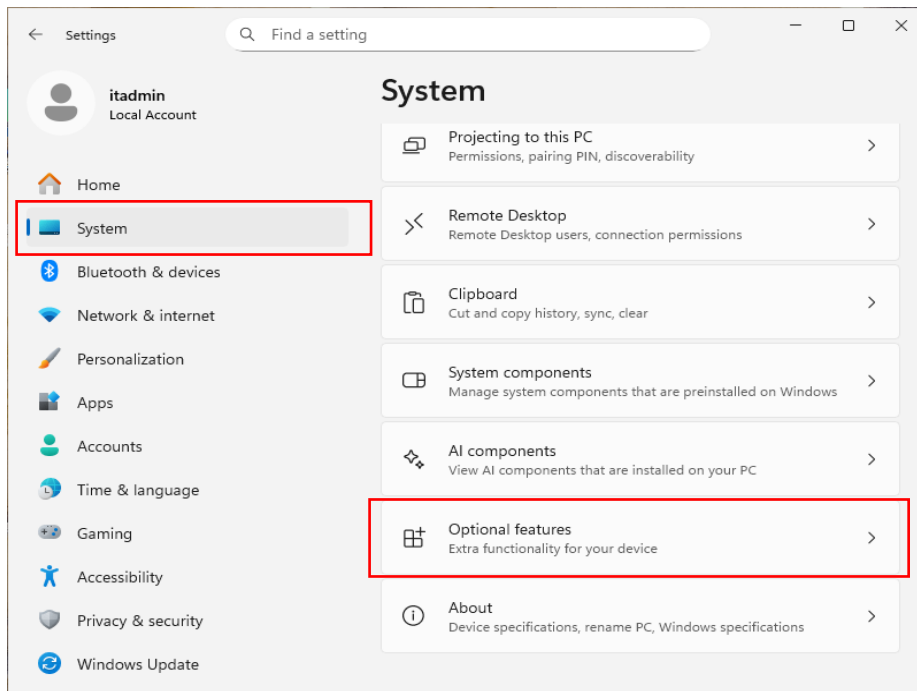
- Select Groups -> Remote Destop Users -> Add -> Enter the username -> click check name -> Enter the credentials -> Click OK



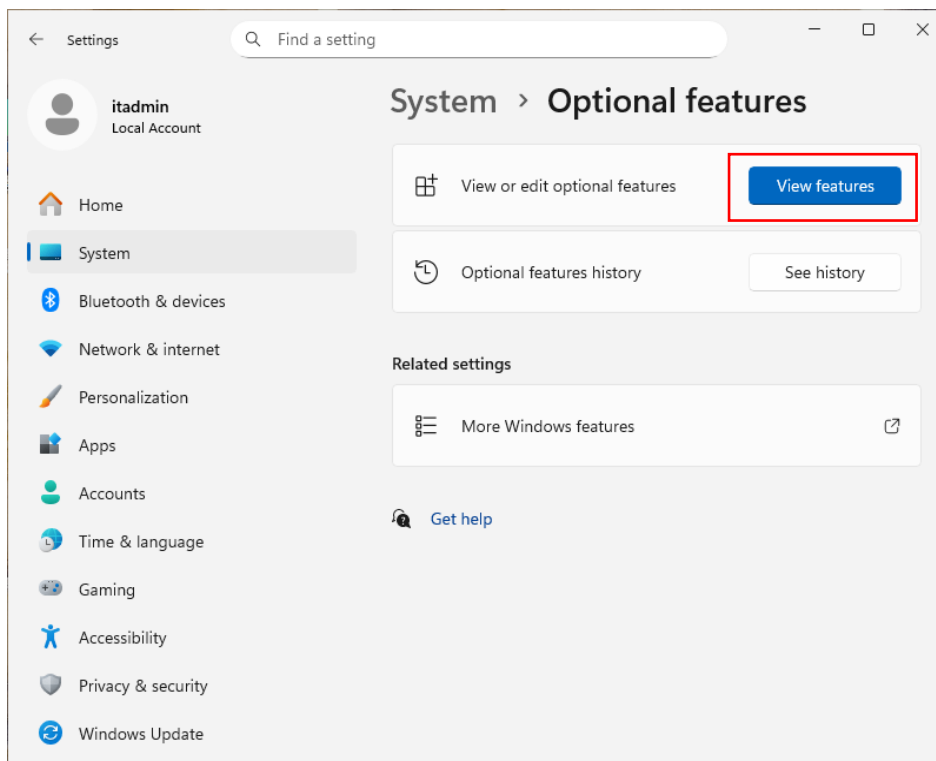
- Again, try login in RDP.

Step-20: Try Creating and Managing user in GUI (RSAT)

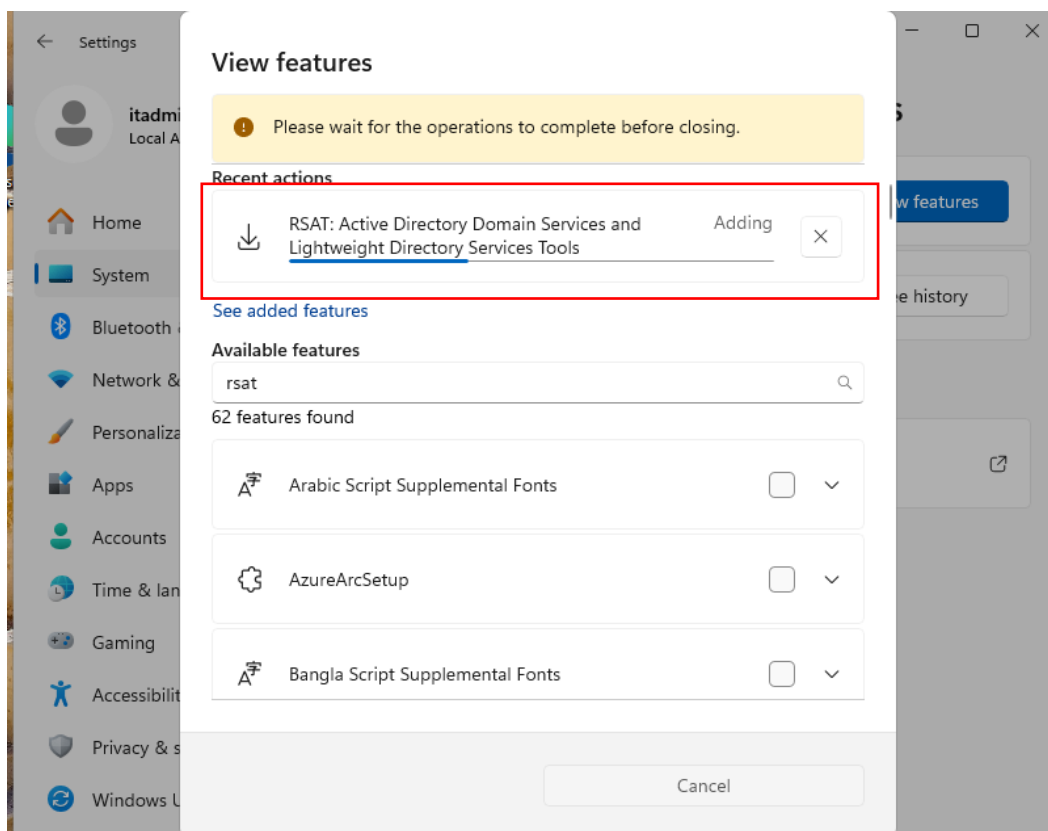
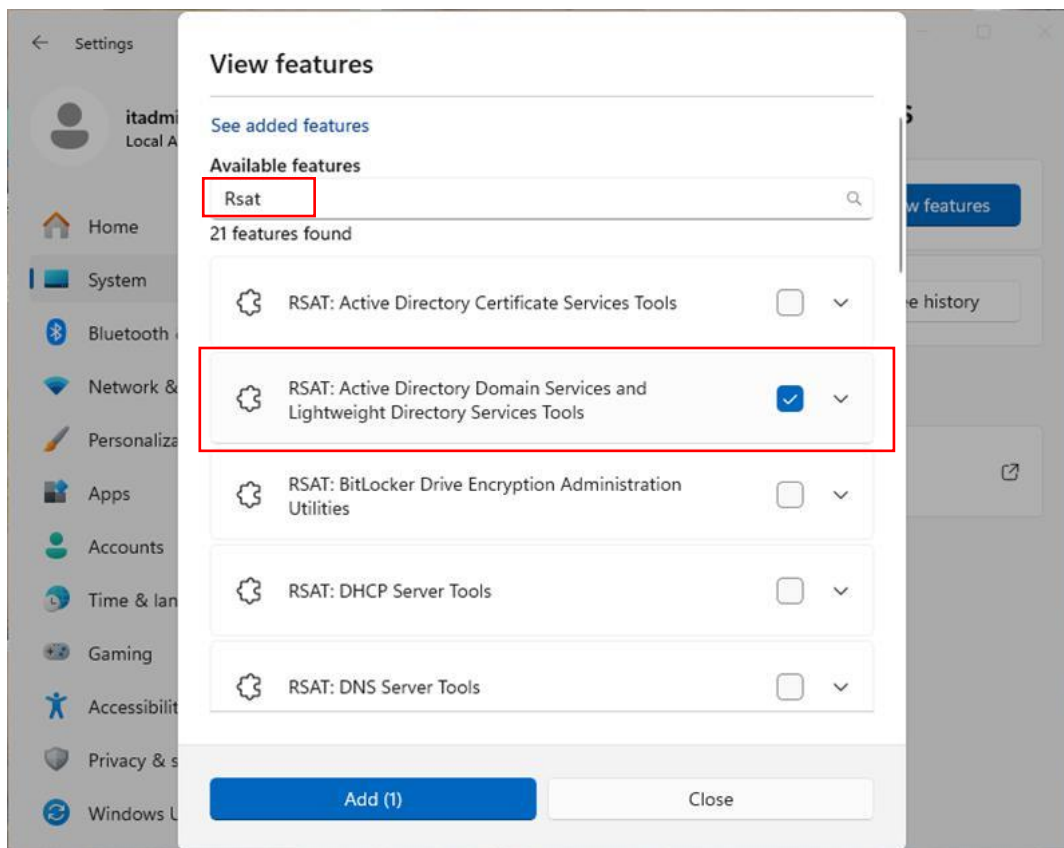
- Login with administrator. Open Windows setting, Select 'system' -> Optional Features



- Click on 'View features'

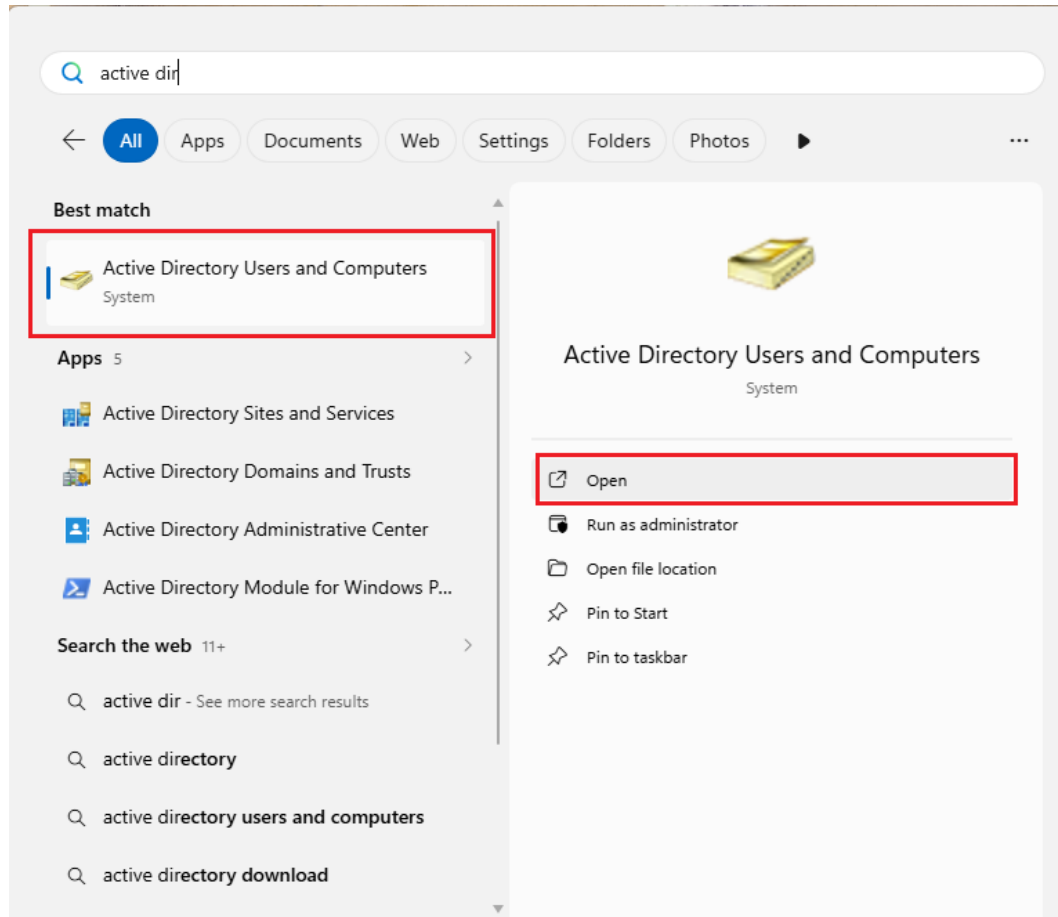


- Select 'see Available features' -> Search for 'Rsat' -> Select 'RSAT: Active Directory services and Lightweight directory service tools' -> Click 'Add' to download.

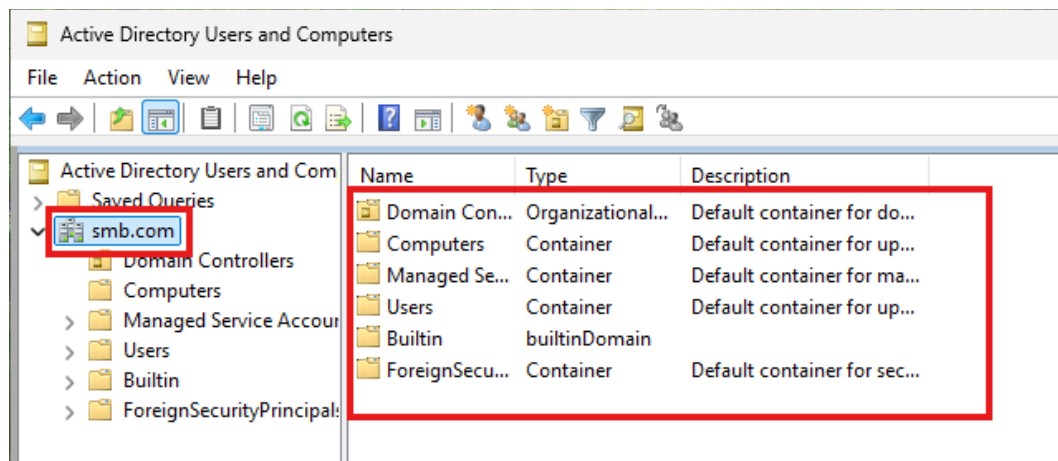


Step-21: To manage the Active Directory Users and Computers

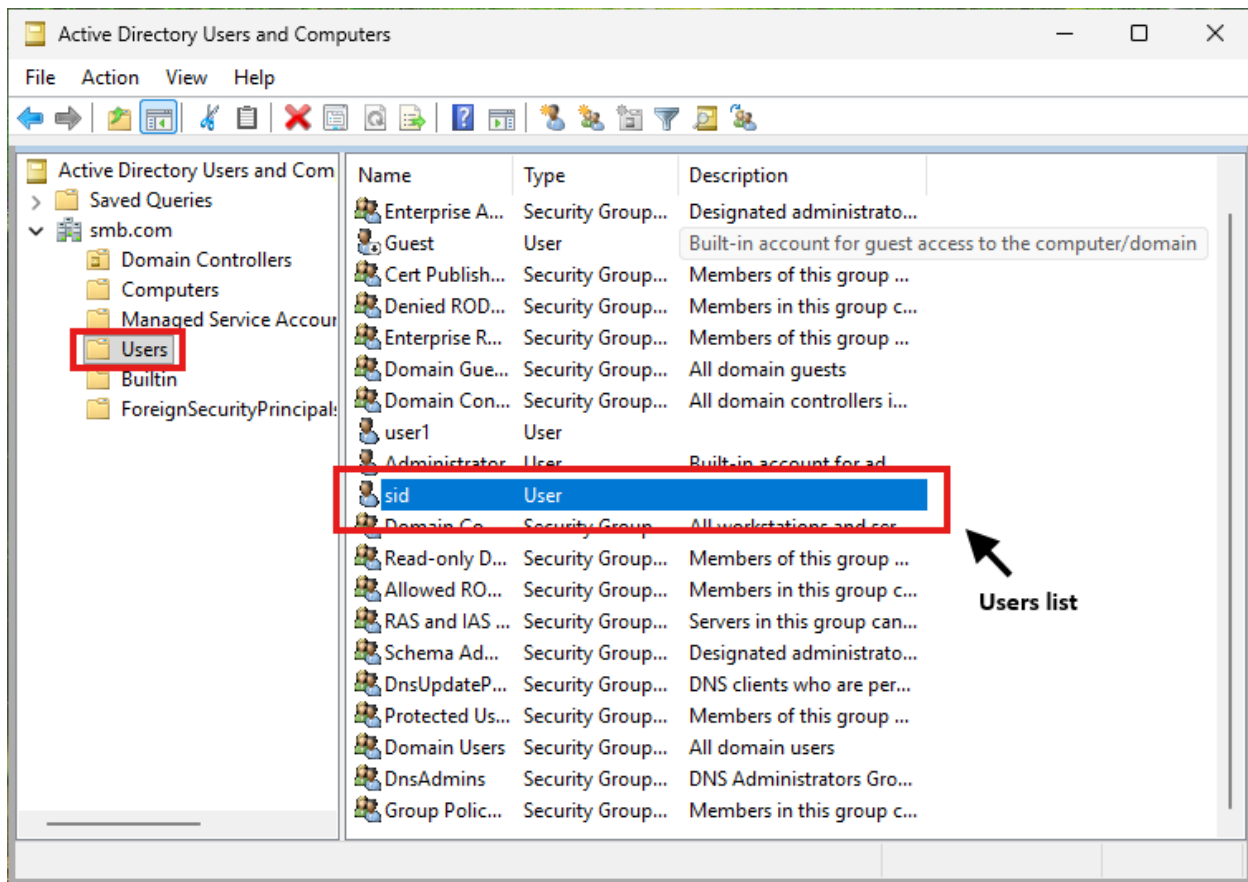
- Once the RSAT is downloaded.
- Press 'win key' and Search 'active directory users and computers' -> Open



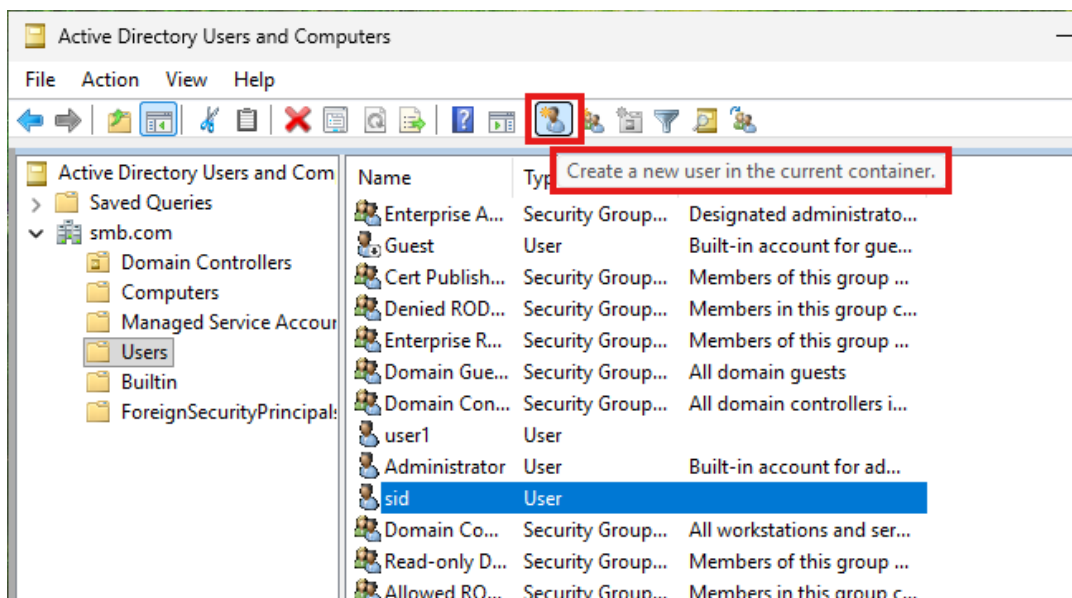
- Click on 'smb.com' (Domain name) -> You can see list of options.



- Click on 'Users' in the sidebar. Users list will be available



- Click on Create users in the top toolbar.



- Enter the First Name, Username, logon name and Password

New Object - User

Create in: smb.com/Users

First name: Siddhu Initials:

Last name:

Full name: Siddhu

User logon name: siddhu@smb.com

User logon name (pre-Windows 2000): SMB\ siddhu

< Back Next > Cancel

New Object - User

Create in: smb.com/Users

Password:

Confirm password:

☐ User must change password at next logon

☐ User cannot change password

☐ Password never expires

☐ Account is disabled

< Back Next > Cancel

New Object - User

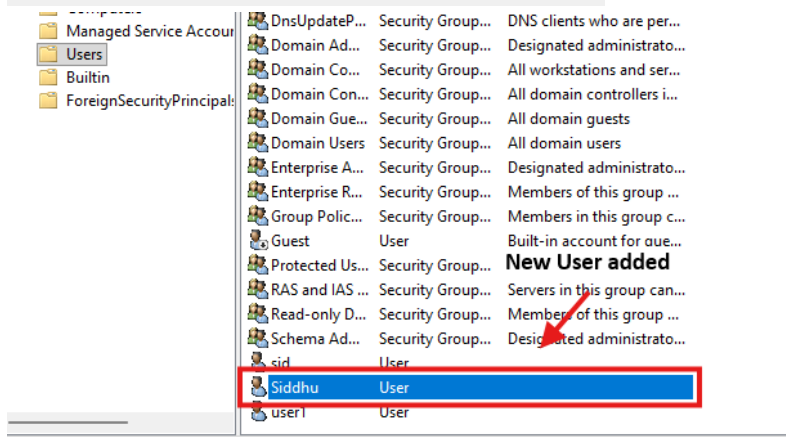
Create in: smb.com/Users

When you click Finish, the following object will be created:

Full name: Siddhu

User logon name: siddhu@smb.com

< Back Finish Cancel



Adding Linux machine (Fedora) to samba AD as client machine

Step-22: Get Hostname and add it to hosts

- Open Linux machine, Get hostname

hostname -f

```
sid@fedora-test:~$ hostname -f  
fedora-test
```

sudo vi /etc/hosts (Ensure the entry)

```
sid@fedora-test:~$ cat /etc/hosts  
# Loopback entries; do not change.  
# For historical reasons, localhost precedes localhost.localdomain:  
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4  
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6  
  
127.0.1.1    fedora-test  
  
# Domain controller  
10.10.40.88  master.smb.com master  
  
# See hosts(5) for proper format and other examples:  
# 192.168.1.10 foo.example.org foo  
# 192.168.1.13 bar.example.org bar
```

← Client IP and hostname

← Admin IP and hostname

Step-23: Install the required packages

sudo dnf install -y sssd realmd adcli krb5-workstation samba-common-tools oddjob oddjob-mkhomedir

```
sid@fedora-test:~$ sudo dnf install -y sssd realmd adcli krb5-workstation samba-common-tools oddjob oddjob-mkhomedir  
Updating and loading repositories:  
Repositories loaded.  
Package "realmd-0.17.1-18.fc43.x86_64" is already installed.  
Package "samba-common-tools-2:4.23.5-2.fc43.x86_64" is already installed.
```

Package	Arch	Version	Repository	Size
Upgrading:				
krb5-libs	x86_64	1.22.2-3.fc43	updates	2.3 MiB
replacing krb5-libs	x86_64	1.21.3-7.fc43	fedora	2.3 MiB
Installing:				
adcli	x86_64	0.9.2-10.fc43	fedora	339.0 KiB
krb5-workstation	x86_64	1.22.2-3.fc43	updates	2.5 MiB
oddjob	x86_64	0.34.7-17.fc43	fedora	138.8 KiB
oddjob-mkhomedir	x86_64	0.34.7-17.fc43	fedora	44.3 KiB
sssd	x86_64	2.12.0-1.fc43	updates	34.3 KiB
Installing dependencies:				
krb5-pkinit	x86_64	1.22.2-3.fc43	updates	125.2 KiB
libipa_hbac	x86_64	2.12.0-1.fc43	updates	61.2 KiB
libkadm5	x86_64	1.22.2-3.fc43	updates	214.0 KiB
sssd-ad	x86_64	2.12.0-1.fc43	updates	416.1 KiB
sssd-common-pac	x86_64	2.12.0-1.fc43	updates	216.8 KiB
sssd-ipa	x86_64	2.12.0-1.fc43	updates	681.7 KiB
sssd-krb5	x86_64	2.12.0-1.fc43	updates	84.9 KiB
sssd-ldap	x86_64	2.12.0-1.fc43	updates	151.4 KiB

Transaction Summary:

Step-24: Modify Resolved

sudo vi /etc/resolv.conf (Ensure the entry)

```
sid@fedora-test:~$ sudo vi /etc/resolv.conf  
nameserver 10.10.40.88  
nameserver 127.0.0.53  
options edns0 trust-ad  
search .
```

Step-25: Modify Krb5

sudo vi /etc/krb5.conf

Add this content: (Ensure the entry)

rdns= false

default_realm = SMB.COM

```
sid@fedora-test:~$ sudo vi /etc/krb5.conf
```

```
includedir /etc/krb5.conf.d/

[logging]
    default = FILE:/var/log/krb5libs.log
    kdc = FILE:/var/log/krb5kdc.log
    admin_server = FILE:/var/log/kadmind.log

[libdefaults]
    dns_lookup_realm = false
    ticket_lifetime = 24h
    renew_lifetime = 7d
    forwardable = true
    rdns = false
    pkinit_anchors = FILE:/etc/pki/tls/certs/ca-bundle.crt
    spake_preauth_groups = edwards25519
    dns_canonicalize_hostname = fallback
    qualify_shortname = ""
    default_realm = SMB.COM ← Domain name
    default_ccache_name = KEYRING:persistent:%{uid}

[realms]
# EXAMPLE.COM = {
#     kdc = kerberos.example.com
#     admin_server = kerberos.example.com
# }

[domain_realm]
# .example.com = EXAMPLE.COM
# example.com = EXAMPLE.COM
```

- Restart the services
sudo systemctl restart kadmin krb5kdc

```
sid@fedora-test:~$ sudo systemctl restart kadmin
sid@fedora-test:~$ sudo systemctl restart krb5kdc
```

Step-26: Discover the domain controller

sudo realm -v discover smb.com

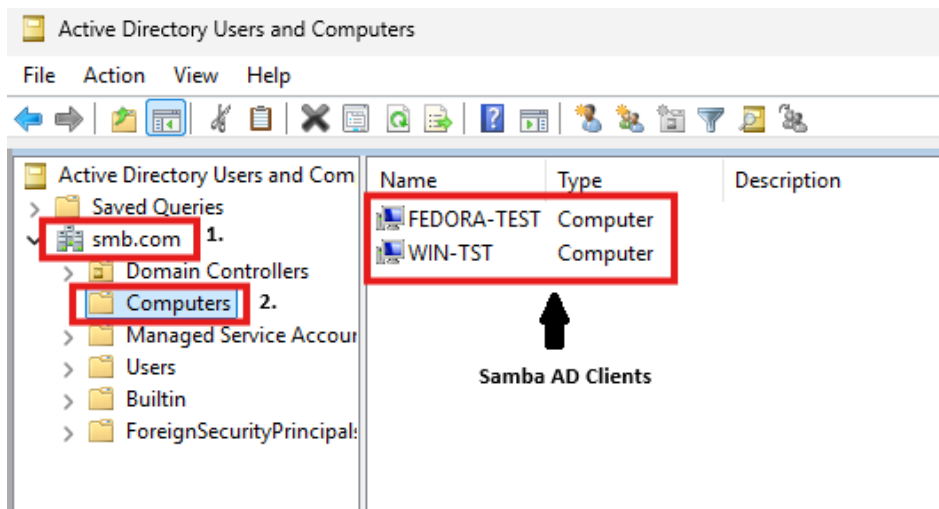
```
sid@fedora-test:~$ sudo realm -v discover smb.com
* Resolving: _ldap._tcp.smb.com
* Performing LDAP DSE lookup on: 10.10.40.88
* Successfully discovered: smb.com
smb.com
  type: kerberos
  realm-name: SMB.COM
  domain-name: smb.com
  configured: no
  server-software: active-directory
  client-software: sssd
  required-package: sssd-common
  required-package: oddjob
  required-package: oddjob-mkhomedir
  required-package: sssd-ad
  required-package: adcli
  required-package: samba-common-tools
```

sudo realm join -U Administrator smb.com

```
sid@fedora-test:~$ sudo realm join -U Administrator smb.com
[sudo] password for sid:
Password for Administrator@SMB.COM:
Warning: Your password will expire in 38 days on Fri 22 May 2026 06:11:33 PM IST
```

Step-27: Verify the Domain controller in windows (RSAT)

- Open Active Directory users and computers
- Select smb.com -> computers -> use can see client's machines



Step-28: Try login domain

sudo login

```
sid@fedora-test:~$ sudo login
Hangup
sudo login
```

if, output like this. Try ↓

su - sid@smb.com -----> user name and domain name

```
sid@fedora-test:~$ su - sid@smb.com
Password:
Creating home directory for sid@smb.com.
sid@smb.com@fedora-test:~$ |
```

```
sid@smb.com@fedora-test:/home$ whoami
sid@smb.com
```

```
sid@smb.com@fedora-test:~$ cd /home/
sid@smb.com@fedora-test:/home$ ll
total 20
drwx-----. 16 sid          sid          4096 Apr 14 14:51 sid
drwx-----.  4 sid@smb.com domain users@smb.com 4096 Apr 14 15:35 sid@smb.com
drwx-----.  3 sri          sri          4096 Apr 14 11:14 sri
drwx-----. 22 thamjeed    thamjeed     4096 Apr 14 11:18 thamjeed
drwx-----. 19 vijaycs     vijaycs      4096 Apr  2 18:18 vijaycs
sid@smb.com@fedora-test:/home$ |
```

Step-29: If you wanted to change the home directory in Linux machine (Fedora)

- Make changes in sssd

sudo vi /etc/sss/sss.conf

- Change the fallback_homedir = /data/home/%u@%d

```
[sssd]
domains = smb.com
config_file_version = 2
services = nss, pam

[domain/smb.com]
default_shell = /bin/bash
krb5_store_password_if_offline = True
cache_credentials = True
krb5_realm = SMB.COM
realmd_tags = manages-system joined-with-adcli
id_provider = ad
fallback_homedir = /data/home/%u@%d
ad_domain = smb.com
use_fully_qualified_names = True
ldap_id_mapping = True
access_provider = ad
```

To change the home directory

- Restart the service

```
# sudo systemctl restart sssd
```

```
sid@fedora-test:~$ sudo systemctl restart sssd
```

- Login user and check the home dir

```
sid@fedora-test:~$ su - user1@smb.com
Password:
Creating home directory for user1@smb.com.
Last login: Tue Apr 14 16:08:24 IST 2026 on pts/7
```

```
user1@smb.com@fedora-test:~$ pwd
/data/home/user1@smb.com
```

Deploying applications to Samba client machines.

Add Group policy to Samba AD DC (Windows)

Step-30: Install Group policy

- Open setting -> optional features -> View features -> Search and select 'RSAT group policy Management Tools' -> Click add

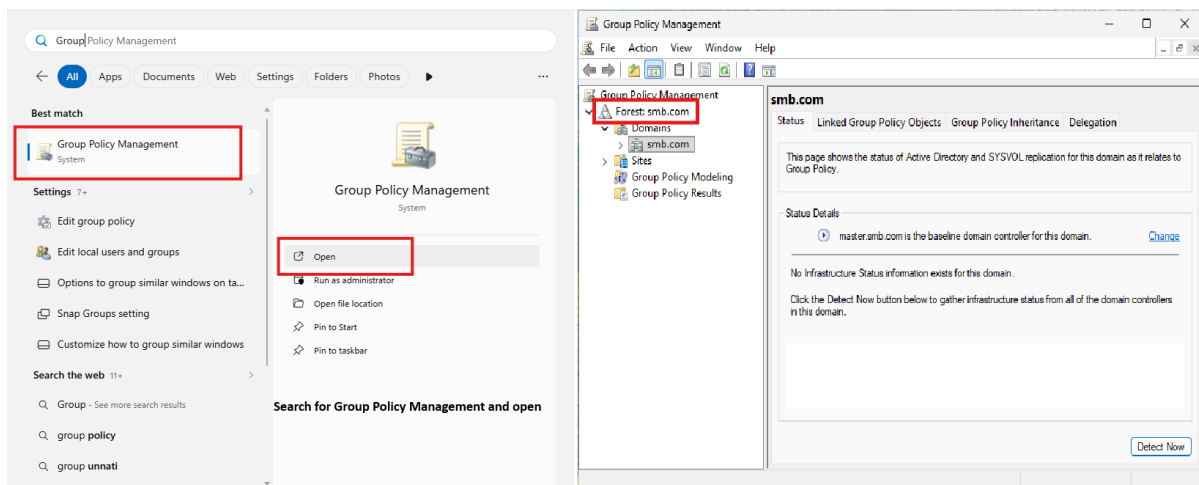
View features

⚠ Please wait for the operations to complete before closing.

Recent actions

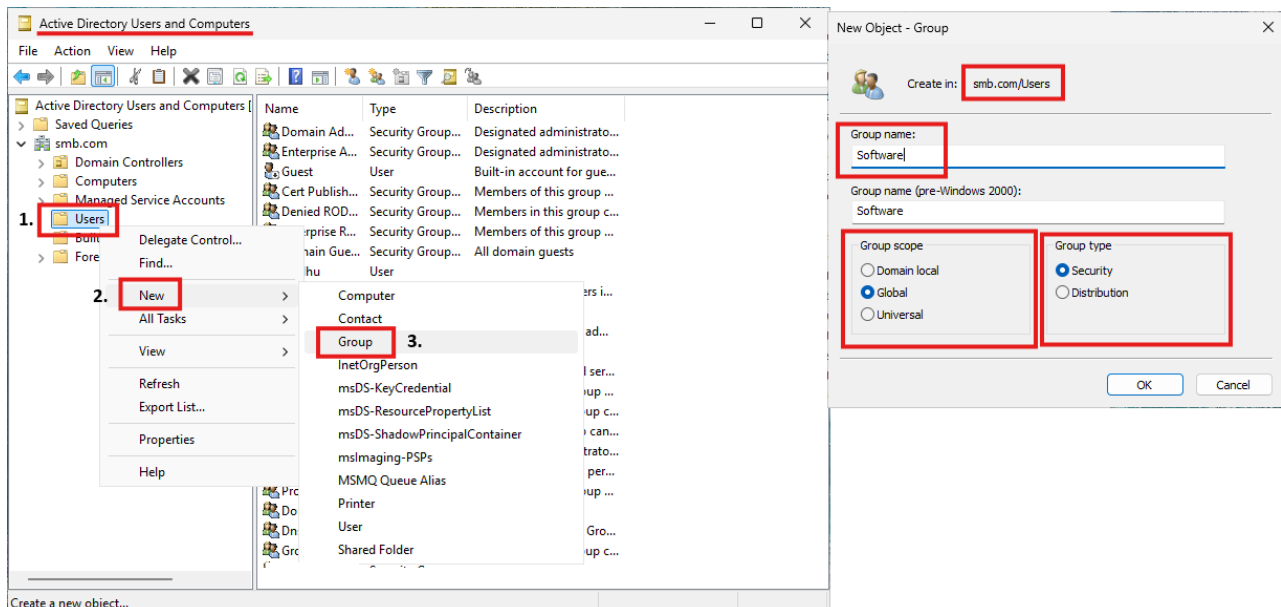
↓ RSAT: Group Policy Management Tools Adding ×

- Once added. Open Group policy management.



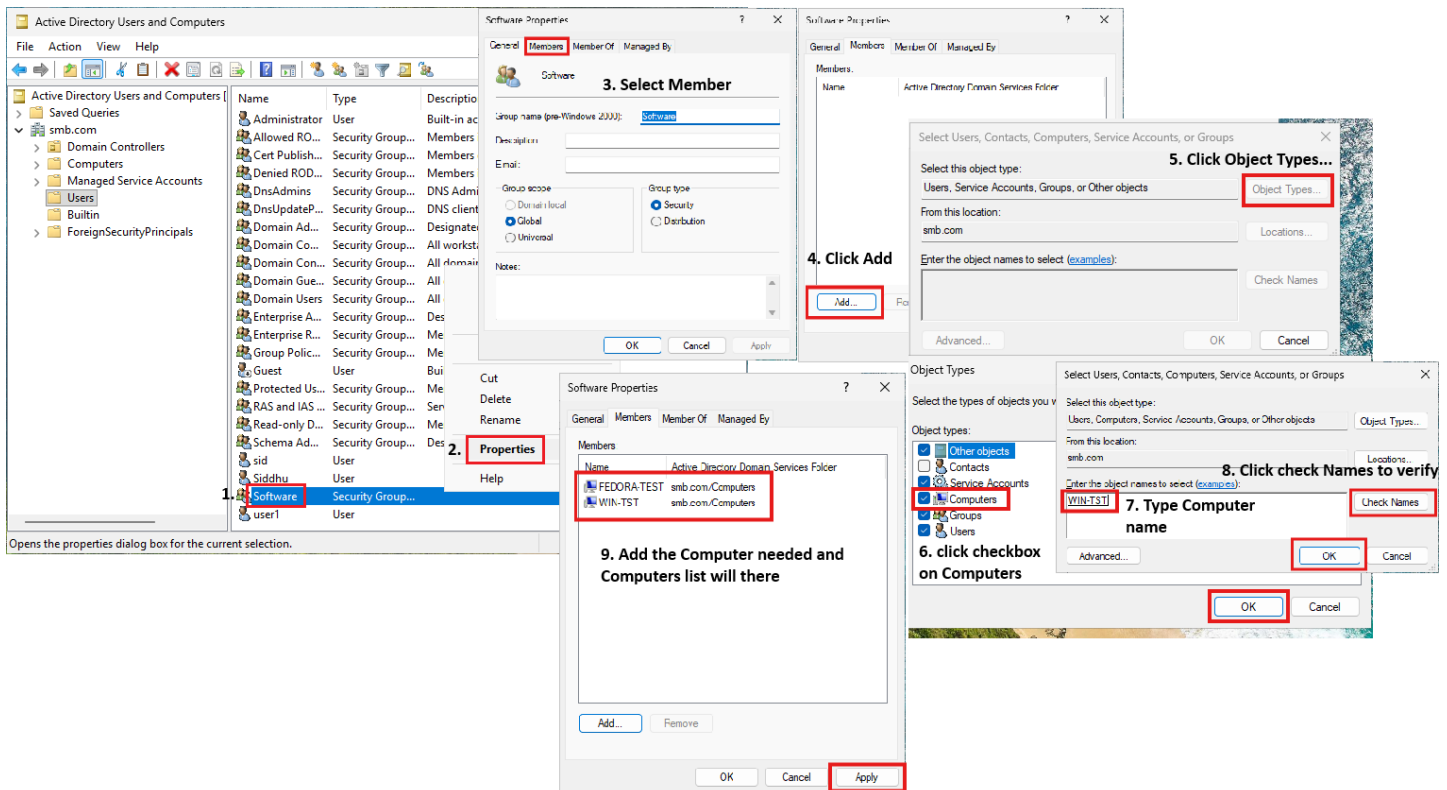
Step-31: First open Active directory users and groups. And create a group

- Press 'win key' -> search 'Active Directory Users and Computers'
- Right click 'Users' -> 'new' -> Click 'Group' -> Enter the Group Name and Change the Group scope and Group type if needed -> Click 'ok'



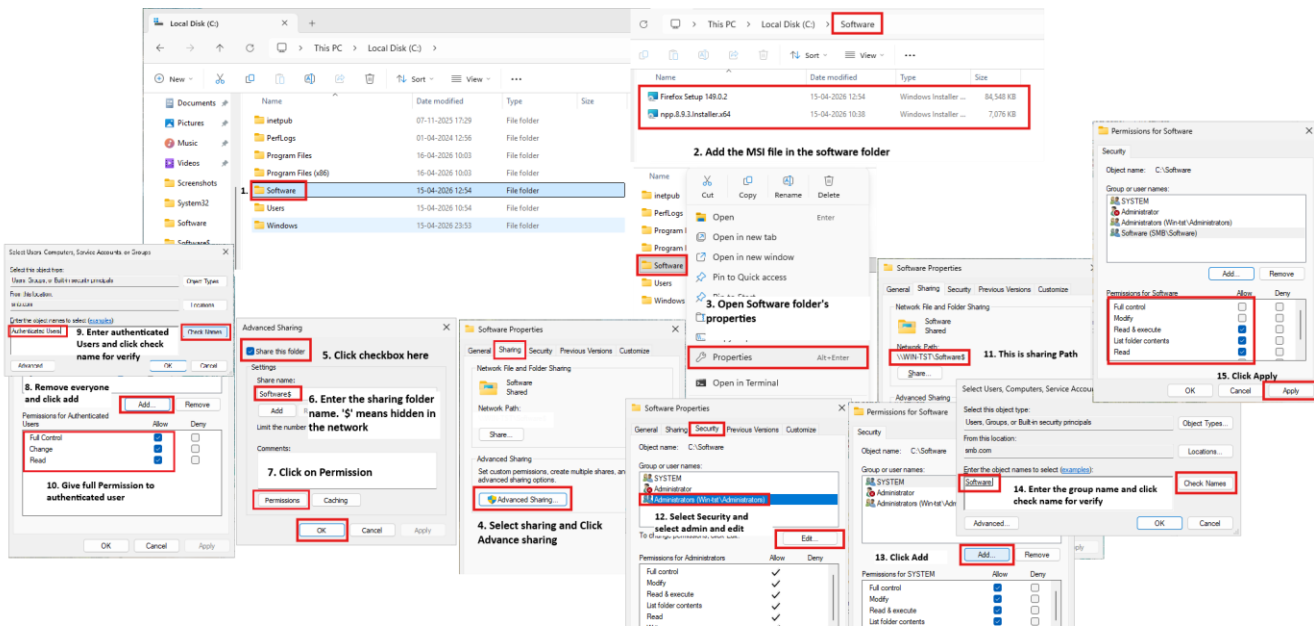
Step-32: Modify the group Properties.

- Find the Created group and right click on group 'Software' -> 'Properties' -> Select 'Members' -> Click 'Add' -> Click 'Object Types...' -> Select 'Computers checkout' -> Enter the Computer name and click 'check Names' -> Click 'ok'



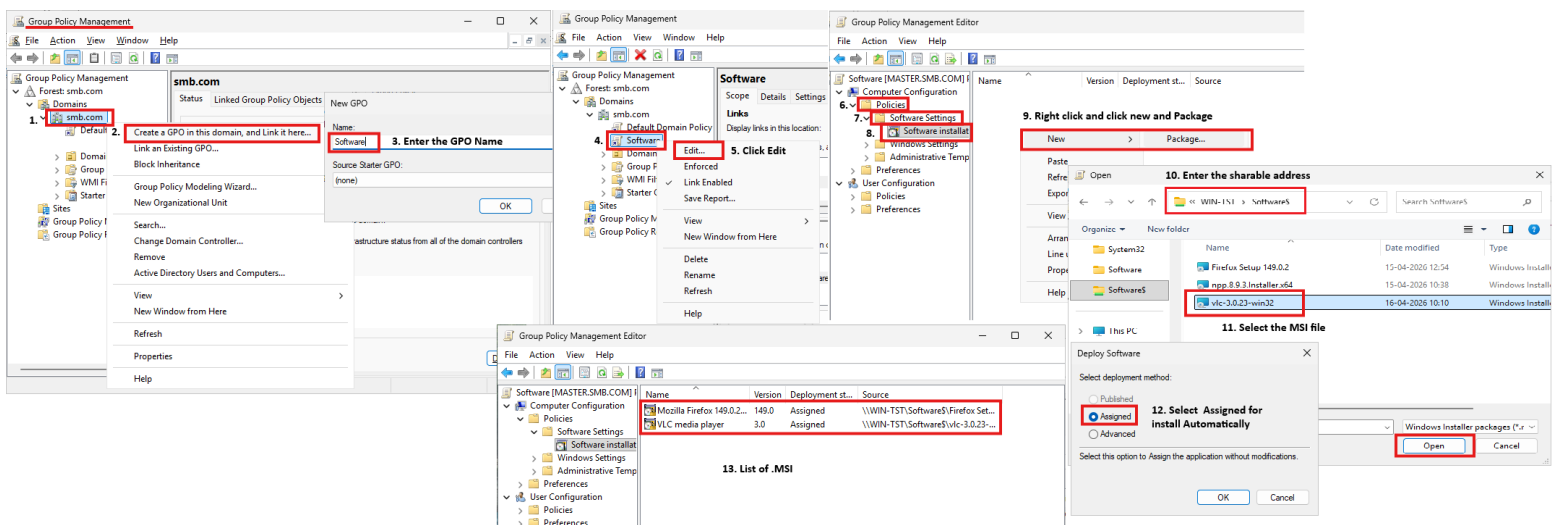
Step-33 : Setup MSI install path and packages

- Open 'File Manager' -> Create a Folder in any location 'Software' -> Download and paste .msi files in this folder.
- Change the Software folder's properties
- Right Click on 'Software' folder -> Click 'Properties' -> Select 'Sharing' in the top -> Click 'Advance Sharing' -> Enable the 'Share the folder' checkbox -> Enter the Share Name 'Software\$' (\$ - hidden from network) -> Click 'Permissions' -> Remove 'Everyone' and Click 'Add' -> Enter 'Authenticated User' -> Click 'Check Names' -> Click 'ok' -> Give full Permission -> Click 'ok'.
- You can find Network Sharing Path. '\\WIN-TST\\Software'
- Select 'Security' -> Select 'Administrators' -> 'Edits...' -> Click 'Add' -> Enter the group name 'Software' -> Click 'Check Name' -> Click 'ok' -> Click 'Apply'.



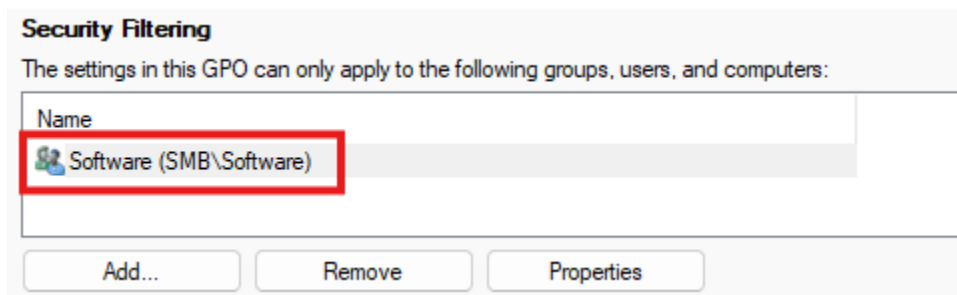
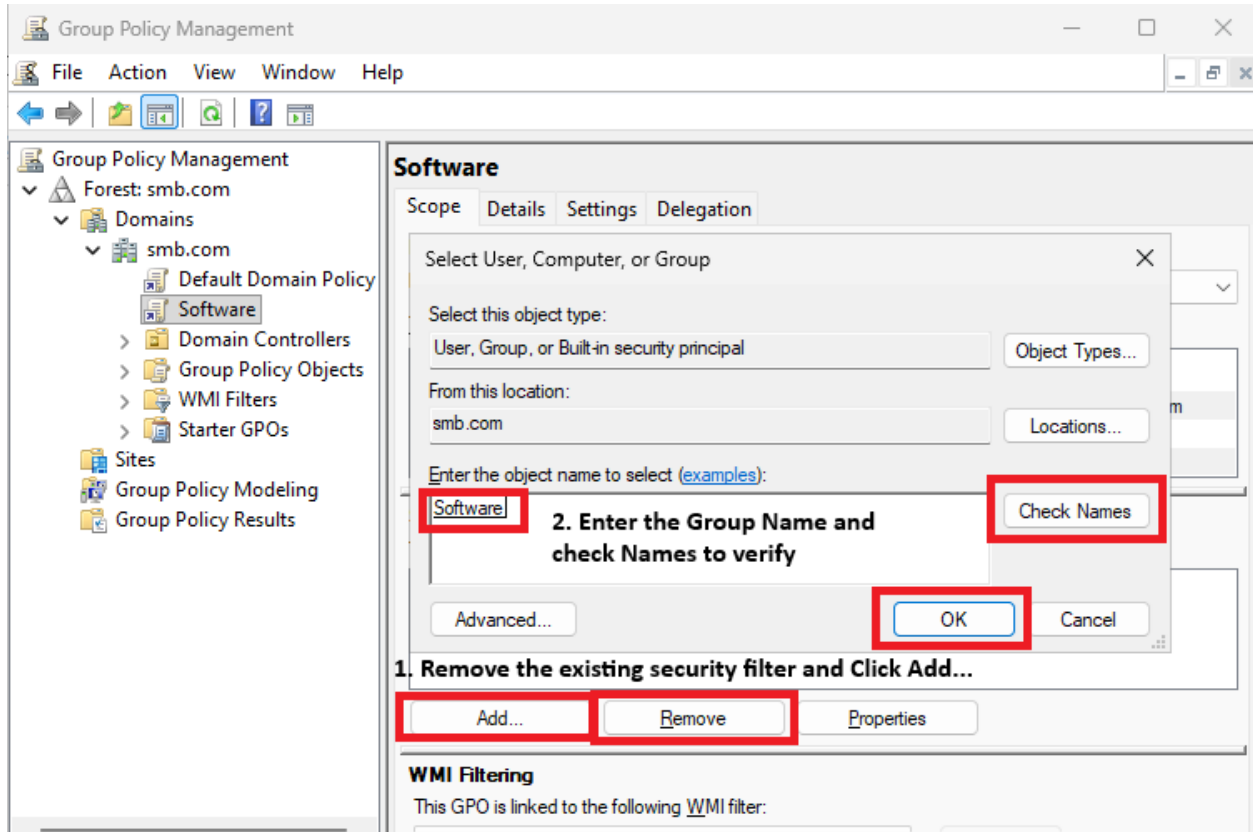
Step-34 : Create Group Policy and add the .msi files

- Open 'Group Policy Management' -> Click 'Create a GPO in this domain, and Link it here' -> Enter GPO Name 'Software'.
- New GPO will be created, Right click on 'Software' -> click 'Edit'.
- Group Policy Management Editor will open. Click Expand 'Policies' -> 'Software Settings' -> Click 'Software Installation' -> Right click on plain area -> 'New' -> 'Package'.
- Select the Application that needs to be installed. Note: use Network Sharing Path. Do not use local machine path.
- Deploy Software prompt appears -> Select 'Assigned' -> click 'ok' -> Application needed will be listed.



Step-35: Change the Security Filter in Software GPO.

- Open 'Group Policy Management' -> Select 'Software' -> Remove the Existing group in security filter.
- Click 'Add...' -> Enter the Group Name 'Software' -> Click 'Check Names' -> Click 'ok'



Step-36: Once completed with all the steps. Update group policy

- Open 'Command Prompt' -> Enter 'gpupdate /force' -> it will ask for restart 'Y'
- Once restarted you can find the application has been installed (It will take bit time to install app)

```
Administrator: Command Prom...
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\administrator.SMB>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.

The following warnings were encountered during computer policy processing:

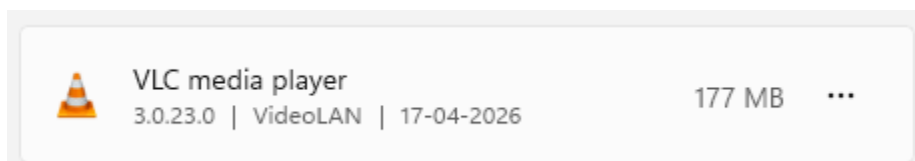
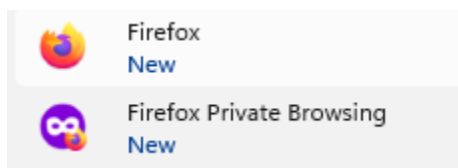
The Group Policy Client Side Extension Software Installation was unable to apply one or more settings because the changes must be processed before system startup or user logon. The system will wait for Group Policy processing to finish completely before the next startup or logon for this user, and this may result in slow startup and boot performance.
User Policy update has completed successfully.

For more detailed information, review the event log or run GPRESULT /H GPREport.html from the command line to access information about Group Policy results.

Certain Computer policies are enabled that can only run during startup.

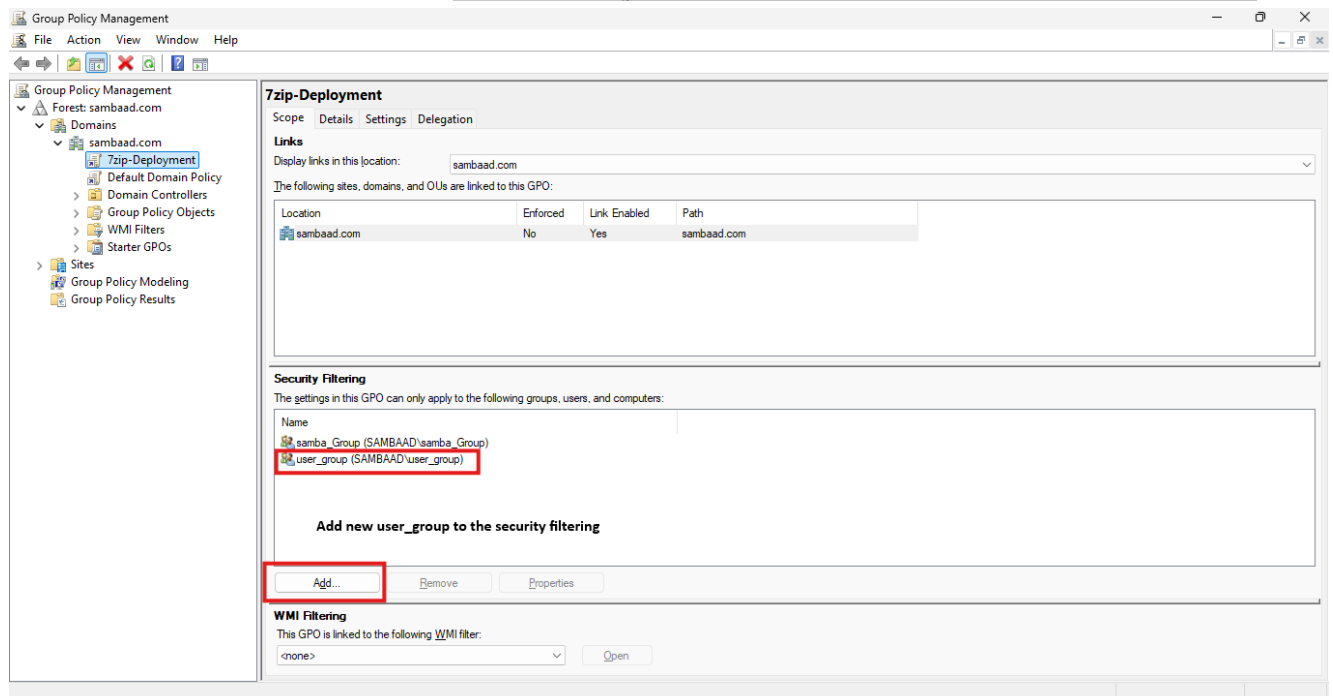
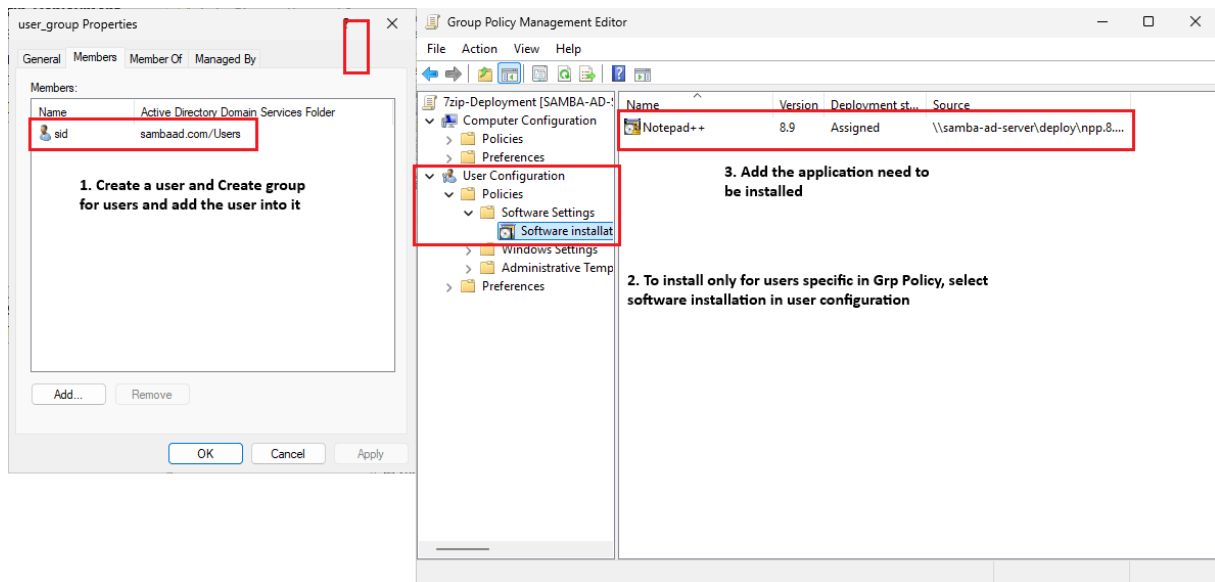
OK to restart? (Y/N)y
```

- New Application Has been added.



Step-37: If you want to add separately to users alone.

- Follow steps 31, 32,34,35,36
- Follow steps as doing for computer group configuration.
- First, create a group -> add users
- Open group policy management -> right click on the GPO -> click edit.
- Group policy Management editor will open in that click user configuration -> Policies -> software settings -> click on software installation -> Right click in plain area -> 'New' -> 'Package' -> select the app (.msi).
- Add user group to security filtering.



Reference:

<https://ubuntu.com/server/docs/how-to/samba/provision-samba-ad-controller/>

https://www.server-world.info/en/note?os=Ubuntu_24.04&p=samba&f=7

https://youtu.be/KFpu3I9vgyl?si=VB_v_jfyCSKO9U6f

<https://youtu.be/HBLIcv3w9IQ?si=oduts34XuN-yq9F0>

<https://youtu.be/KT6O-TfJ41g?si=CSZNT9FV8plAUhSF>

<https://youtu.be/nm3A4opJq2I?si=3dUKQWc76pj2lXgS>

<https://www.youtube.com/watch?v=tDpGM8SQYP4>